# UHPPOTE Desk

## User manual

Web application to manage UHPPOTE UT0311-L0x access controllers: doors, cards, permissions, time profiles, real-time monitoring, event history and attendance.

Access control · Live monitor · Time profiles · Attendance · Multi-operator

EN · IT · ES · FR · DE

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github.com/alexbn71/UHPPOTE-Desk

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### About this manual

The screenshots were taken from a demonstration installation with one simulated 4-door controller and invented people. Your names, numbers and dates will differ. Installation (Docker, network, environment variables) is covered in a separate guide, `docs/DOCKER.md` in the project repository.


### Acknowledgements

UHPPOTE Desk is built on the protocol knowledge published by the open-source **uhppoted** project (codeberg.org/uhppoted). UHPPOTE Desk is an independent project, not affiliated with uhppoted or with the controller manufacturer.

PART 1

# Understanding UHPPOTE Desk

What the application does, how it talks to the access controllers and the few ideas you need before using it.

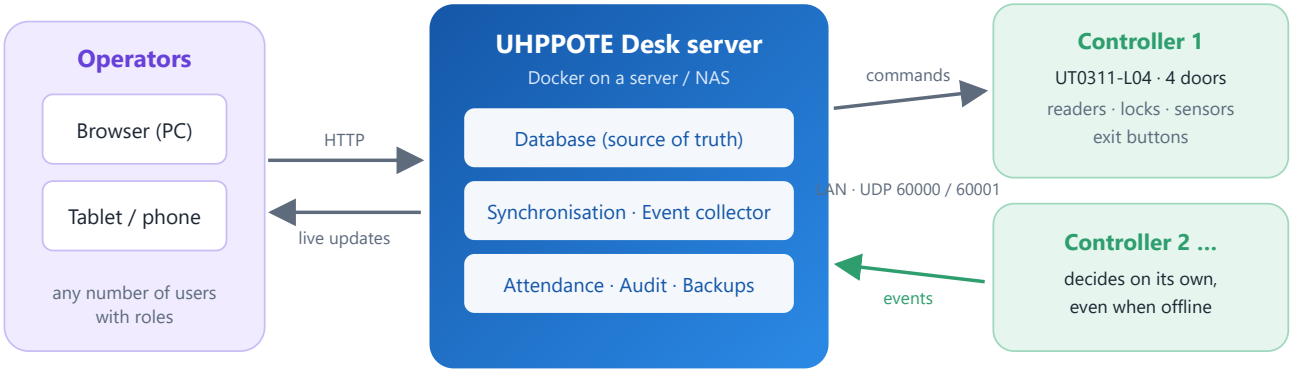
## 1 · What UHPPOTE Desk is

UHPPOTE Desk is a **web application** to manage UHPPOTE **UT0311-L0x** access controllers (1, 2 or 4 doors; this manual uses the 4-door UT0311-L04). It replaces the Windows program supplied with the controllers ("Access Control", N3000) with a modern application that:

- runs on a server or NAS (normally as a Docker container) and is used from any **browser** — PC, tablet or phone — without installing anything;
- is used by **several operators at the same time**, each with a role (administrator, operator, read-only) and, if needed, restricted to some areas of the building;
- keeps people, cards and permissions in its own **database** and automatically sends every change to the controllers;
- receives the swipes and alarms **in real time**, keeps the full **event history** and computes **attendance** (hours worked, late arrivals, absences);
- is available in **English, Italian, Spanish, French and German**; each user chooses the language.

## 2 · How the system works

An access control installation is made of three parts: the **controllers** mounted near the doors, the **server** running UHPPOTE Desk and the **browsers** of the operators.




- **The controllers decide.** When a card is swiped, the controller itself checks whether that card may open that door at that time, and opens the lock. It does not ask the server, so doors keep working even if the server or the network is down.

- **The server is the reference.** People, cards, permissions, time profiles and settings are kept in the UHPPOTE Desk database. Every change made in the application is sent straight away to the controllers that are online. A controller that was offline is brought up to date with a Synchronisation.
- **Events flow back.** Every swipe, door opening or alarm is recorded in the controller's memory and sent to the server. The server stores it in the event history and shows it live. If the network was down, the missing events are downloaded from the controller as soon as it is back: none is lost.

> **Why this matters**
> Do not change the controllers with other programs (the original Windows software or the controller's web page) while UHPPOTE Desk manages them: the application would undo those changes at the next synchronisation, because its database is the reference.

## 3 · Basic concepts

### Controller

The electronic board that manages 1, 2 or 4 doors. It is identified by its **serial number** (9 digits on the S/N label) and reached over the LAN by its IP address. It stores up to tens of thousands of cards and events.

### Door

Each controller output: a card **reader**, an electric **lock** (relay), optionally a **door sensor** (open/closed) and an **exit button**. In the application a door has a name, an operating mode and an opening time.

### Person

Someone who passes through the doors: employee code, name, department, validity period, optional keypad PIN, photo and attendance shift.

### Card

A proximity card or key fob. Its **number** has 8 digits in Wiegand-26 format: facility code × 100000 + card code, for example `10058400` = facility 100, code 58400. A card can be assigned to a person, or registered as free.

### Permission

The right of a person to open a door: **always**, or only within a **time profile**. Permissions belong to the person and apply to all of their cards.

### Time profile

A time window (days of the week, up to 3 time intervals, validity dates) numbered 2–254. It can be linked to another profile to extend it, e.g. weekdays + Saturday morning.

### Event

Anything the controller records: card swipes (granted or denied, with the reason), door opened/closed, exit button, alarms, power on, remote opening.

### Synchronisation

Aligning a controller with the database. It happens automatically for every change; the Synchronisation page checks and repairs differences (e.g. after a long offline period).

### Zone and department

**Zones** group controllers (buildings, floors) and can restrict what an operator may command. **Departments** organise people and filter attendance reports.

### Attendance

Swipes on readers flagged for attendance are used as clock-in/clock-out punches. With shifts, holidays and leave the application computes hours worked, overtime and late arrivals.

### How an access is decided

When a card is swiped on a door, the controller grants access only if **all** of these are true; otherwise it denies access and records the reason (see Event reasons):

1. the card is stored on the controller (a card of an inactive person, or a lost or disabled card, is removed from the controllers);
2. today is within the validity period of the card holder;
3. the person has a permission on that door — always, or with a time profile that includes the current day and time;
4. the door rules allow it: door mode, interlock, anti-passback, first card, and the PIN when the reader has a keypad.

## 4 · Operators and roles

Everybody who uses the application logs in with their own **operator** account. Each operator has one of three roles. Administrators can also restrict an operator to some **zones**: remote door opening and synchronisation are then allowed only on the controllers of those zones.

| What the operator can do | Read-only | Operator | Administrator |
|---|---|---|---|
| See dashboard, monitor, event history, attendance reports, people, cards, permissions, controllers, doors, time profiles, tasks, system information | ✓ | ✓ | ✓ |
| Export events and attendance to CSV | ✓ | ✓ | ✓ |
| Create and edit people, cards, photos, permissions; import people from CSV; departments | — | ✓ | ✓ |
| Open doors remotely | — | ✓ | ✓ |
| Guest links; plates of people | — | ✓ | ✓ |
| Attendance: shifts, holidays, leave, manual punches | — | ✓ | ✓ |
| Compare and align cards, synchronise the time, download events | — | ✓ | ✓ |
| Controllers (add, network, advanced settings, reset), doors, time profiles, scheduled tasks, zones, full synchronisation | — | — | ✓ |
| Operators, audit log, backups | — | — | ✓ |
| Rules, notifications, webhooks, API tokens, mobile app settings, plate cameras | — | — | ✓ |
| Two-factor authentication on their own account | ✓ | ✓ | ✓ |

Pages and buttons that a role cannot use are hidden or disabled. Every change and every command sent to a controller is recorded in the Audit log with the name of the operator.

## 5 · The user interface

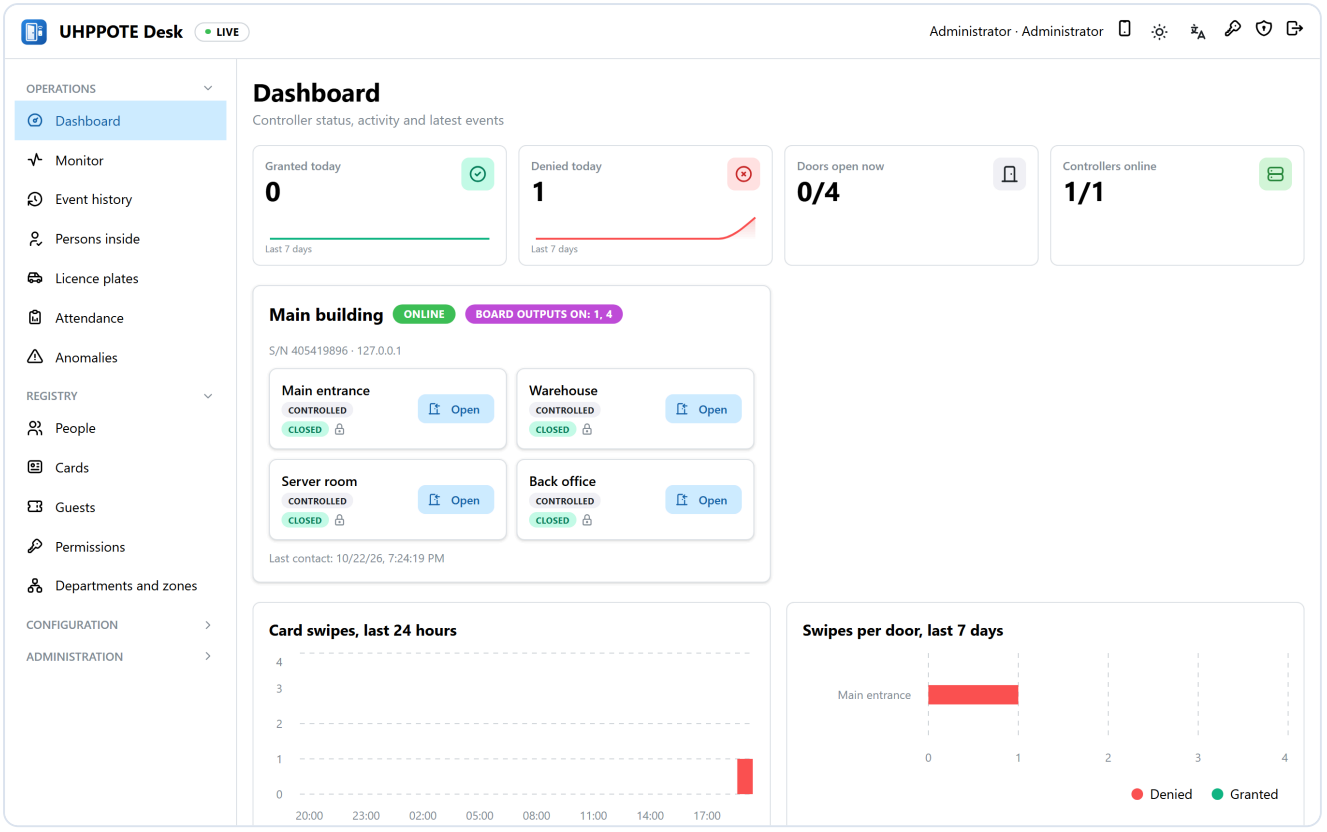



**Figure 1** — The main window: header at the top, navigation on the left, the selected page on the right.

### Header

- **LIVE badge** — green when the browser receives real-time updates from the server (events, door states, synchronisation progress). It turns red, **Offline**, if the connection to the server is lost; it reconnects by itself. It refers to the connection between your browser and the server, not to the controllers.
- **User name and role** of the logged-in operator.
- **Language** ( 文A icon) — English, Italiano, Español, Français, Deutsch. The choice is remembered by the browser; on the first visit the browser language is used.
- **Change password** (key icon) and **Log out**.

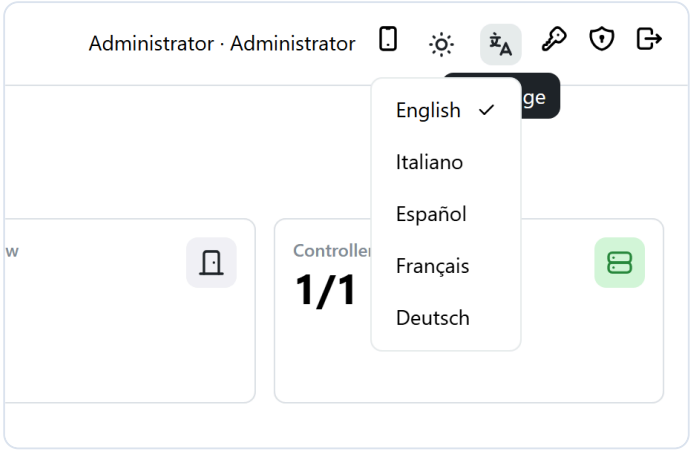



**Figure 2** — The language menu.

### Navigation

The menu on the left is divided into four groups, which are also the parts of this manual:

| Group | Pages | Used for |
|---|---|---|
| OPERATIONS | Dashboard, Monitor, Event history, Attendance | day-to-day supervision |
| REGISTRY | People, Cards, Permissions, Departments and zones | who can go where |
| CONFIGURATION | Controllers, Doors, Time profiles, Scheduled tasks, Synchronisation | hardware and rules |
| ADMINISTRATION | Operators, Audit log, System | users and maintenance |

Messages appear in the top-right corner: green for completed operations, red for errors. A yellow message **Saved, but not sent to every controller** means the change is in the database but some controller was unreachable; it will be aligned at the next synchronisation.

## 6 · Getting started

### Logging in

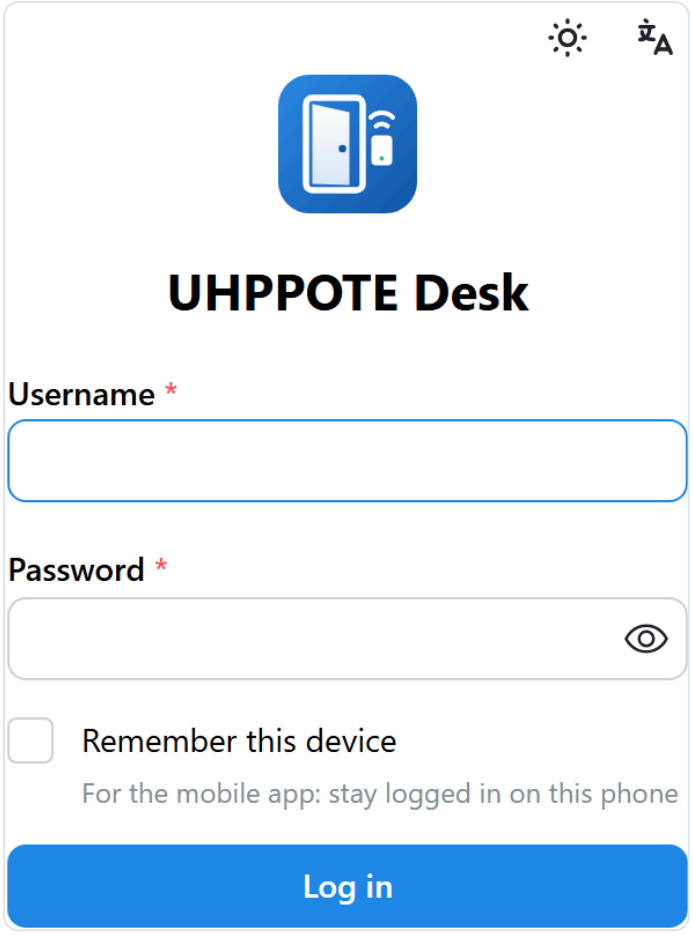



**Figure 3** — The login page.

Open the address of the server in the browser (for example `http://192.168.1.10:8080`) and log in. The first administrator is `admin`:

- if the server was installed with the `ADMIN_PASSWORD` variable, use that password. It can be changed only by changing the variable and restarting the server;
- otherwise a temporary password was generated on the first start and written in the server log. You are asked to choose a new password immediately.

After several wrong passwords the account is blocked for a few minutes. Sessions expire after some hours of use (12 by default).

## First configuration — checklist

1. **Controllers** → **Search network**: find the controllers, add them and give them a name (chapter 17). If a controller has a factory IP address outside your network, change it from its page.
2. **Doors**: give each door a meaningful name, check mode and opening time, and flag the readers to be used for attendance (chapter 18).
3. **Time profiles**: create the time windows you need, e.g. "Office hours" (chapter 19). Permissions valid "always" do not need one.
4. **Departments and zones**: create the organisation of the company and, optionally, the zones of the building (chapter 15).
5. **People**: create them one by one or import a CSV file; assign cards and permissions (chapter 12). You can read the number of a new card simply by swiping it on any reader.
6. **Operators**: create an account for every user of the application with the right role (chapter 22).
7. Check on the **Monitor** that swipes appear in real time. If they do not, see Troubleshooting.

> **Tip**
> Before the hardware arrives you can try everything with the controller simulator included in the Docker image (see `docs/DOCKER.md`, "Trying the app without a controller").

PART 2

# Operations

Day-to-day supervision: controller status, live swipes, event history and attendance.

OPERATIONS

## 7 · Dashboard

The first page after logging in. For every controller it shows a panel with:

- the name, the **Online**/**Offline** state, serial number and IP address, and the time of the last contact. If the controller's clock is wrong by more than a few seconds the drift is shown (the time is corrected automatically);
- a tile for every active door with its **mode** (Controlled, Normally open, Normally closed), the **door sensor** state (Open/Closed) and the **lock** icon, orange when the relay is active (lock released);
- the **Open** button to open the door remotely (operators and administrators): after the confirmation the lock is released for the opening time of the door, and a "Remote opening" event is recorded;
- red badges for **Fire alarm** or **Forced lock** when those inputs of the controller are active.

Below, **Latest events** lists the most recent events of all controllers; denied accesses are highlighted in red. **Open monitor** goes to the live monitor.

The dashboard of Figure 1 shows a typical day: a remote opening, a door opened and closed, an exit button and a card that is not registered (*Card 10065001 — No access rights*).

OPERATIONS

# 8 · Monitor

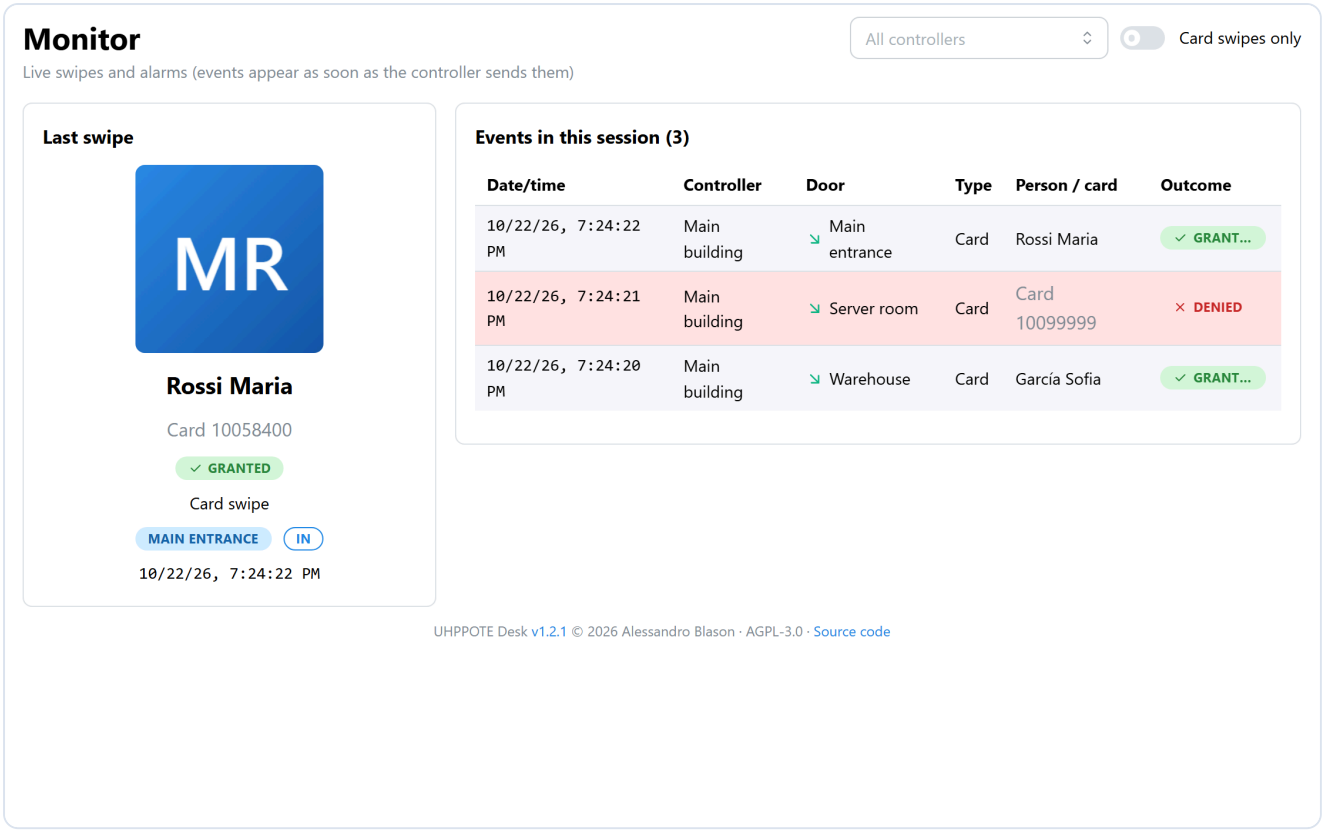



**Figure 4** — Live monitor: the last swipe with the photo of the person, and the events of the session.

The monitor shows swipes and alarms the moment the controller sends them — the equivalent of the "Monitor" of the original software, for example for a reception desk.

- **Last swipe**: photo (or initials), name and card of the person, outcome (**GRANTED** / **DENIED**), reason, door and direction (In/Out), time. An unregistered card is shown as *Unregistered card*.
- **Events in this session**: every event received since the page was opened (up to the most recent 300), newest first.
- Filters at the top: a single controller, and **Card swipes only** to hide door and alarm events.

> **Note**
> The monitor shows only what arrives while it is open. Everything is stored in any case: older events are in the Event history. If swipes do not appear here but do appear later in the history, the server is not receiving events in real time — see Troubleshooting.

OPERATIONS

# 9 · Event history

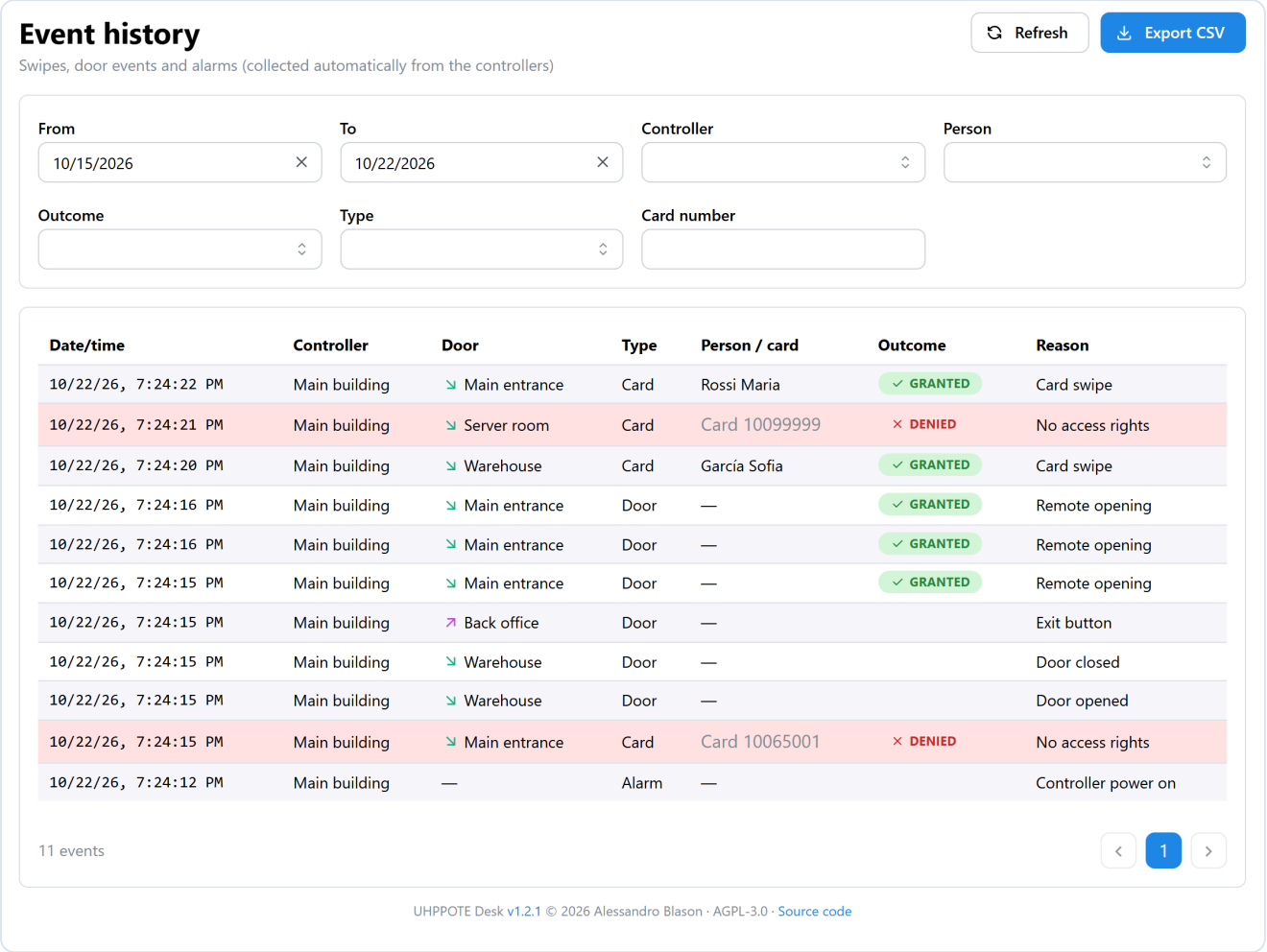



**Figure 5** — Event history with filters.

All the events collected from the controllers, newest first. Use the filters to find what you need:

- **From / To** — period;
- **Controller** and **Door**;
- **Person** or **Card number**;
- **Outcome** — granted or denied only.

Each row shows date and time (controller time), controller, door with direction arrow (↘ in, ↗ out), type (Card, Door, Alarm), person or card, outcome, reason and the index of the event in the controller memory. **Export CSV** downloads the filtered events in the current language, ready for Excel.

> **Where events come from**
>
> Events arrive in real time from the controllers and are also downloaded automatically every few minutes and whenever a controller comes back online, so the history is complete even after network or server outages. The controller memory is circular: if a controller stays disconnected so long that its memory fills up, the oldest events are overwritten.

## Persons inside

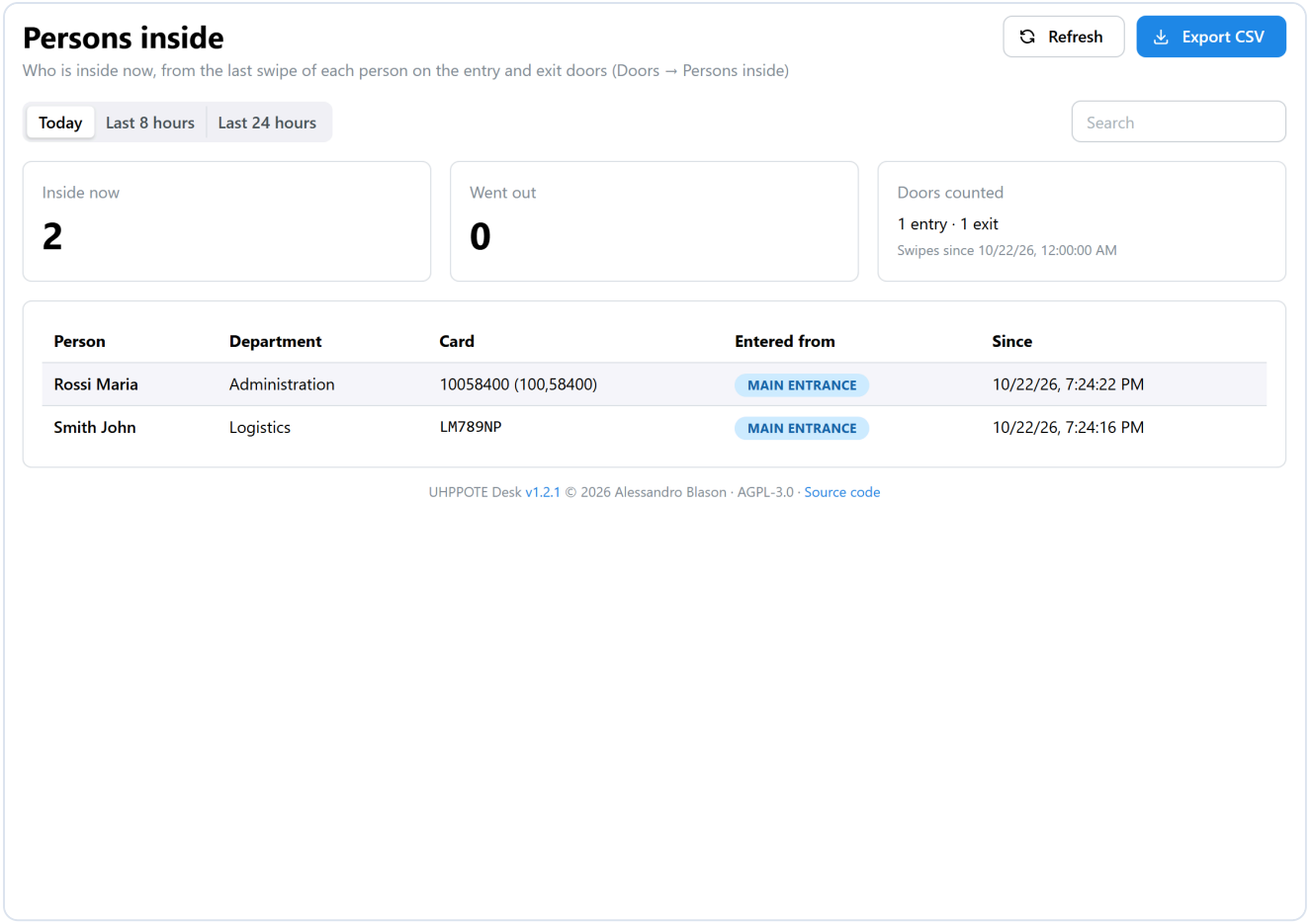



**Figure 6** — Who is inside now.

The Persons inside page shows who is inside now: for every person, the last **granted** card swipe on a door that counts for presence. A swipe on an **entry** door puts the person inside; a swipe on an **exit** door (or on the exit reader of a two-reader door) puts them outside. Set the role of each door on the Doors page: *Entry* (default), *Exit* or *Not counted*.

Choose the period (today, last 8 or 24 hours): people whose last swipe is older are not counted. The page refreshes by itself every 30 seconds; Export CSV downloads the list, for example for a roll call during an evacuation. People leaving with the exit button (without a card) cannot be seen: use a reader on the exit side for exact counts.

OPERATIONS

# 10 · Attendance

UHPPOTE Desk computes working hours from the swipes on the readers flagged as **attendance readers** (on the Doors page) — typically the main entrance. Only **granted** swipes count. The page has five tabs.

## Report

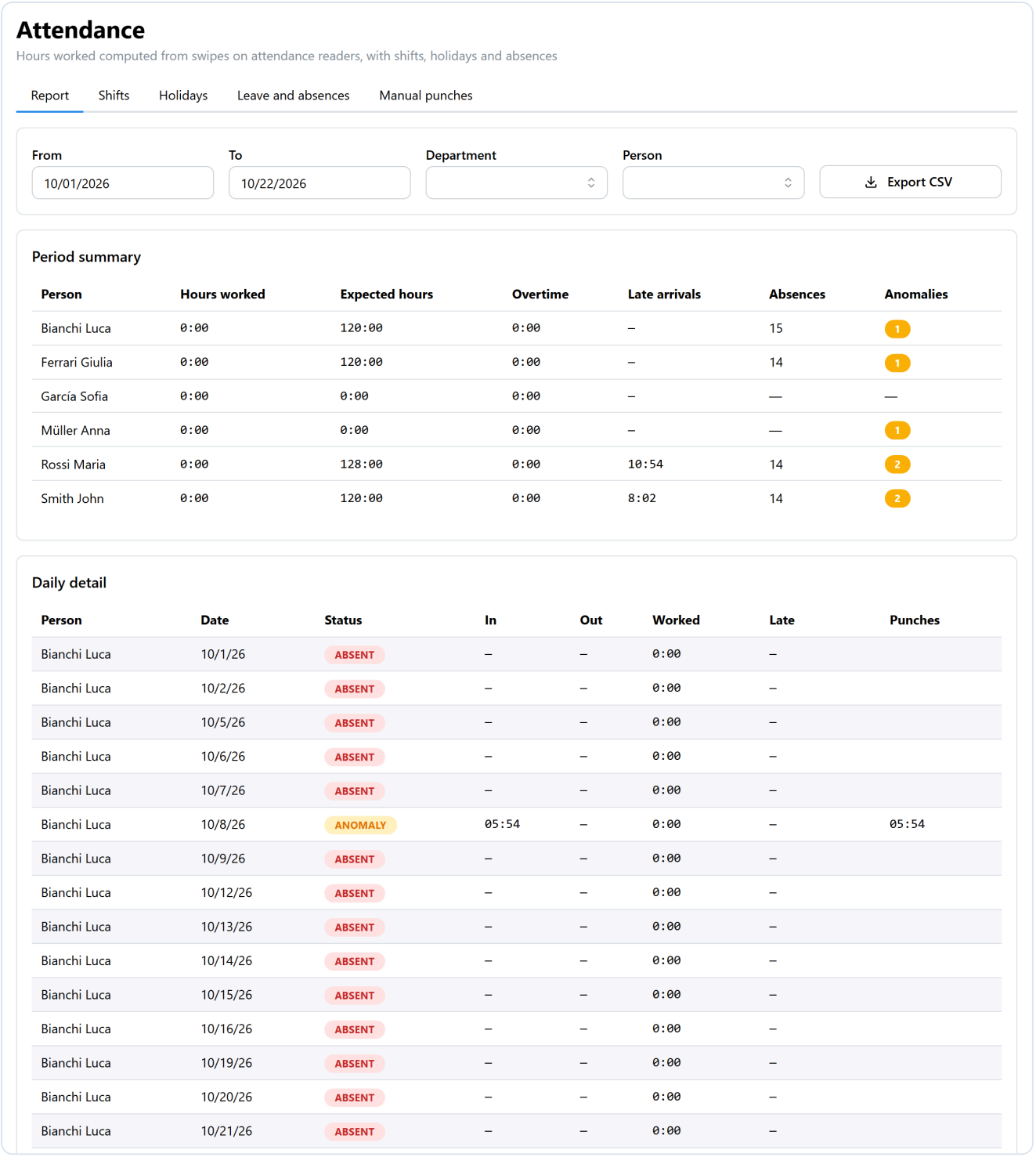



**Figure 7** — Attendance report: period summary and daily detail.

Choose the period (from the first of the current month to today by default), and optionally a department or a person. The **Period summary** shows, per person: hours worked, expected hours, overtime, total late arrivals, days of absence and days with anomalies. The **Daily detail** shows every day with its status (Present, Absent, Public holiday, Day off, Holiday, Permit, Sick leave, Business trip, Anomaly), first in, last out, hours worked, lateness and all the punches; punches entered manually are marked with *****. Export CSV downloads the detail.

### How hours are computed

- Punches are paired in chronological order: in, out, in, out… The direction of the reader is not needed.
- With only one pair of punches in a day, the **break** of the shift is deducted automatically; with two or more pairs the actual break is used.
- An odd number of punches (e.g. a forgotten exit swipe) is an **anomaly**: the last punch is ignored until it is corrected with a manual punch.
- Lateness and early exit take into account the **tolerance** of the shift. Time beyond the expected hours is overtime; on days off and holidays all the time worked is overtime.
- Night shifts (ending after midnight) are attributed to the day on which they started.
- A person without a shift has no expected hours: only the hours worked are computed.

## Shifts

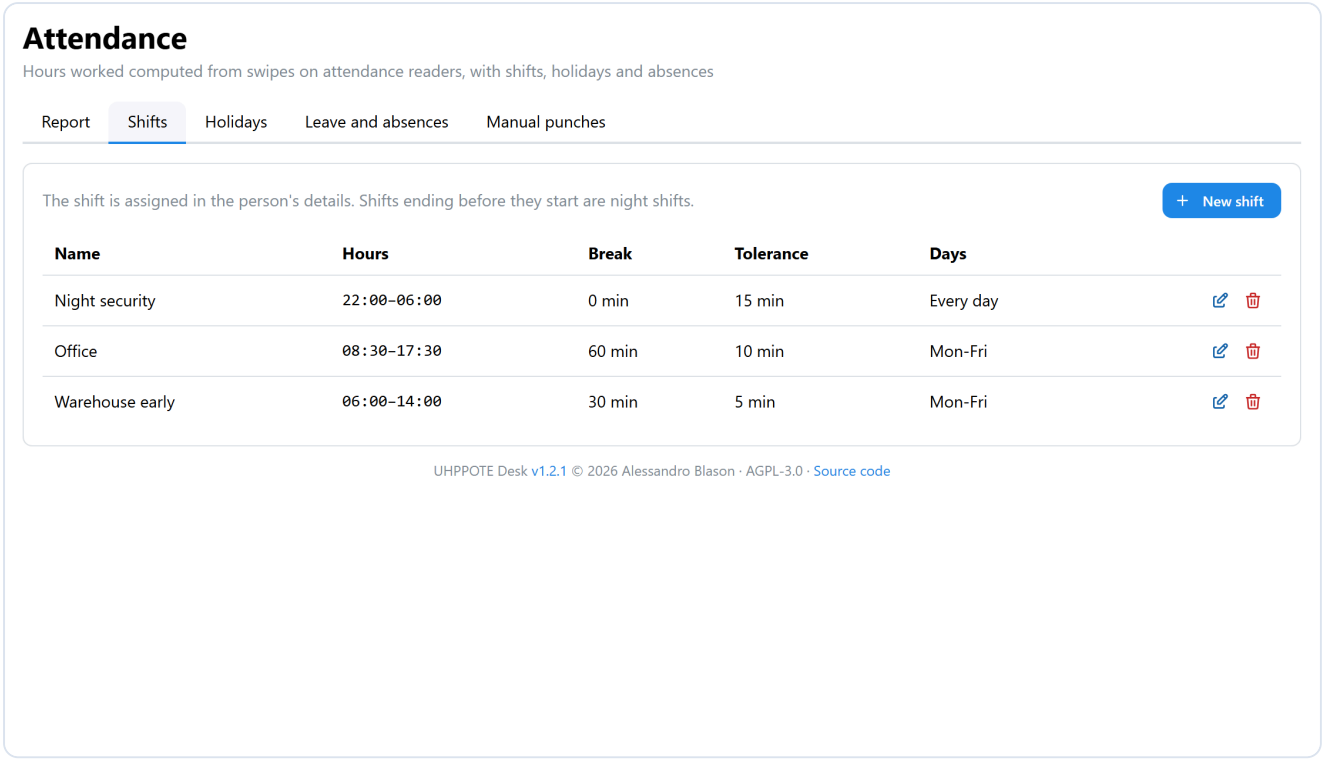



**Figure 8** — Work shifts.

A shift defines start and end time, break (minutes), tolerance (minutes) and working days. A shift that ends before it starts (e.g. 22:00–06:00) is a night shift. Assign the shift in the details of each person (People).

## Holidays

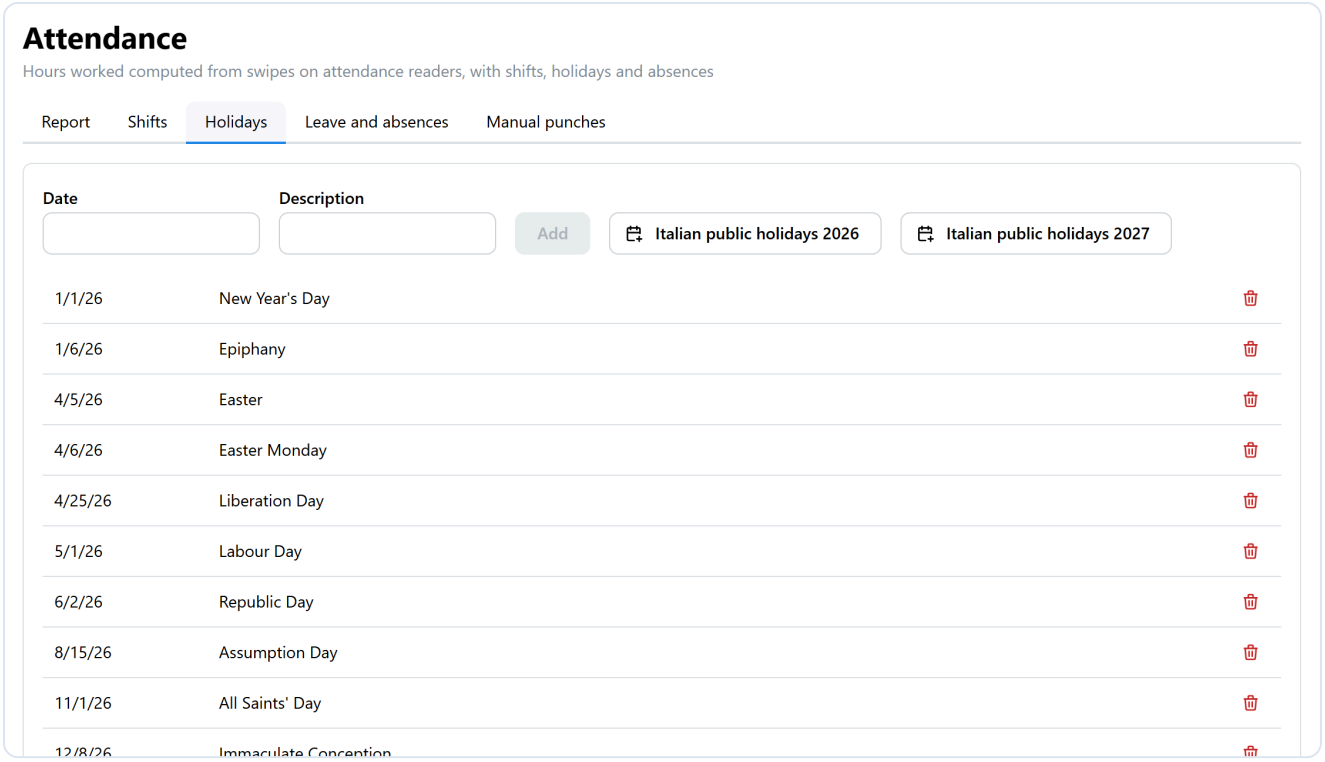



**Figure 9** — Public holidays.

Days on which nobody is expected to work. Add them one by one (date and description) or import the Italian public holidays of a year with one click, Easter included. Add your local holidays, company closures and bridges as needed.

## Leave and absences

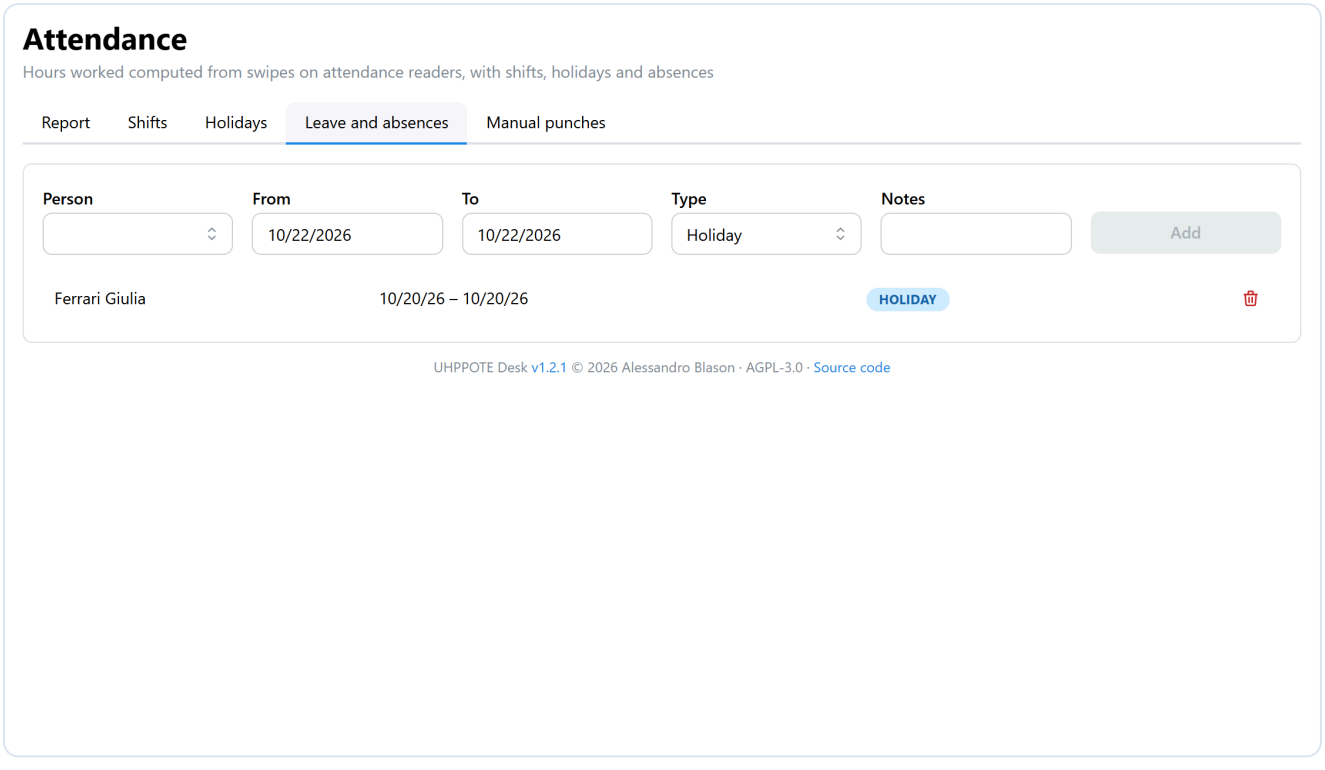



**Figure 10** — Leave and absences.

Record holidays, permits, sick leave, business trips and other absences for a person and a range of dates. Those days are no longer counted as absences in the report.

## Manual punches

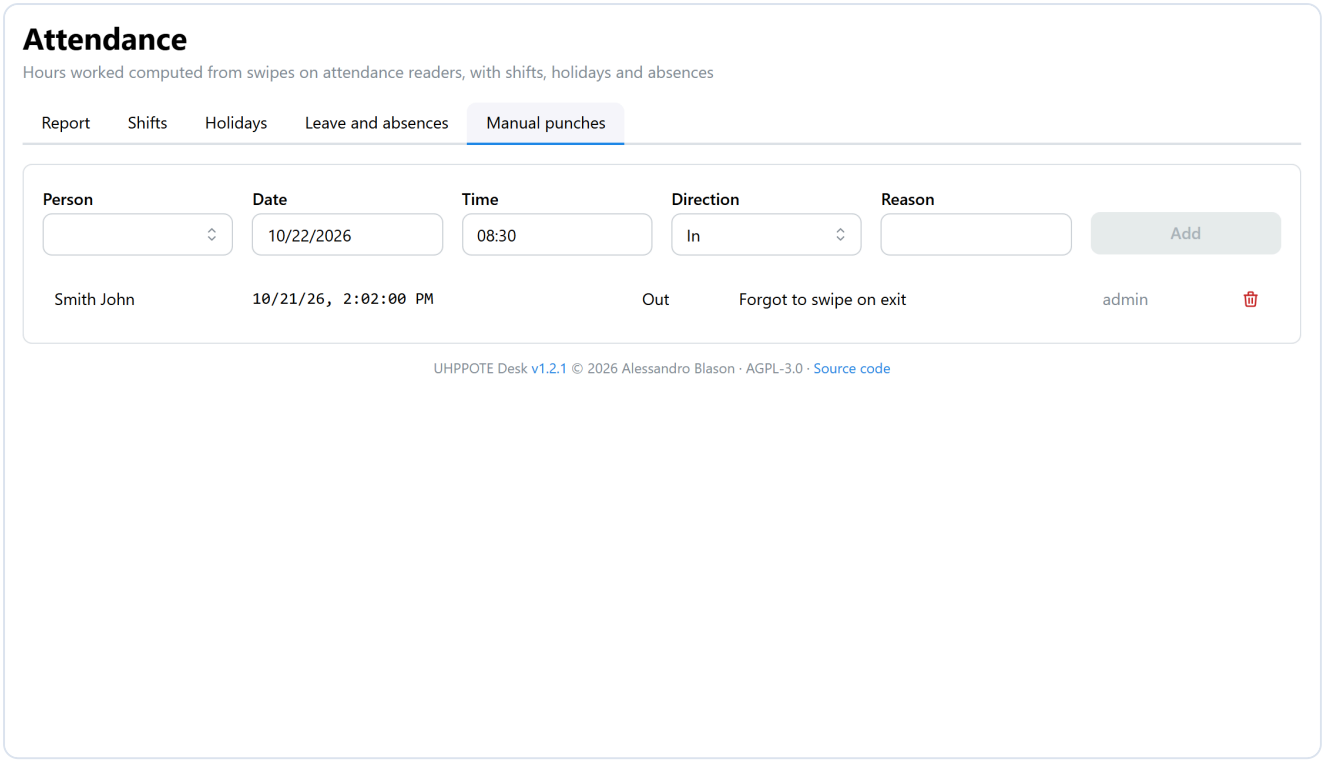



**Figure 11** — Manual punches.

Add a punch that the reader did not record — for example a forgotten exit swipe — with person, date and time, direction and a reason. Manual punches are marked with * in the report and recorded in the audit log.

OPERATIONS

# 11 · Licence plates

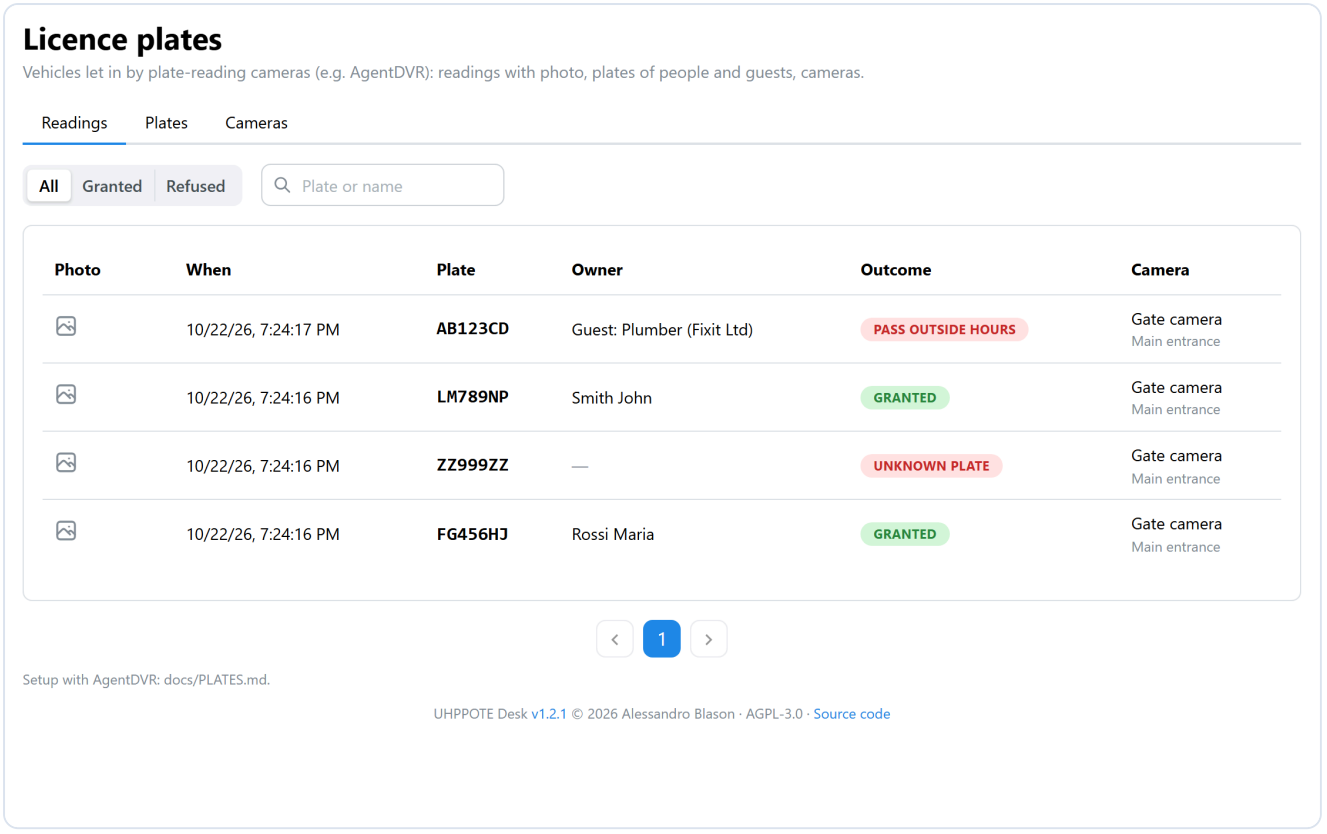



**Figure 12** — Plate readings: who was let in, who was refused.

A camera with plate recognition (tested with **AgentDVR**) sends every plate it reads to UHPPOTE Desk, which decides whether the vehicle may enter, opens the door, takes a photo from the camera and records the reading. The page has three tabs.

- **Readings** — every plate read, with photo, time, owner, outcome (*Granted*, *Unknown plate*, *Outside the time profile*…) and camera. Filter by outcome or search by plate or name. Photos are kept for 30 days by default ( `PLATE_PHOTO_DAYS` ).
- **Plates** — the plates known to the system. A plate belongs to a **person** (added in the person's *Plates* tab: the vehicle passes where the person has a permission, within its time profile) or to a **guest link** (field *Vehicle plates*: the vehicle passes while the link is valid). Plates are compared ignoring spaces, dashes and the characters readers confuse most (B/8, O/0, I/1, Z/2, S/5).
- **Cameras** (administrators) — one row per camera: the door it opens, the photo address, the pause after an opening (the car is still in front of the camera) and *Open the door* (off = readings are only recorded, to test a new camera).

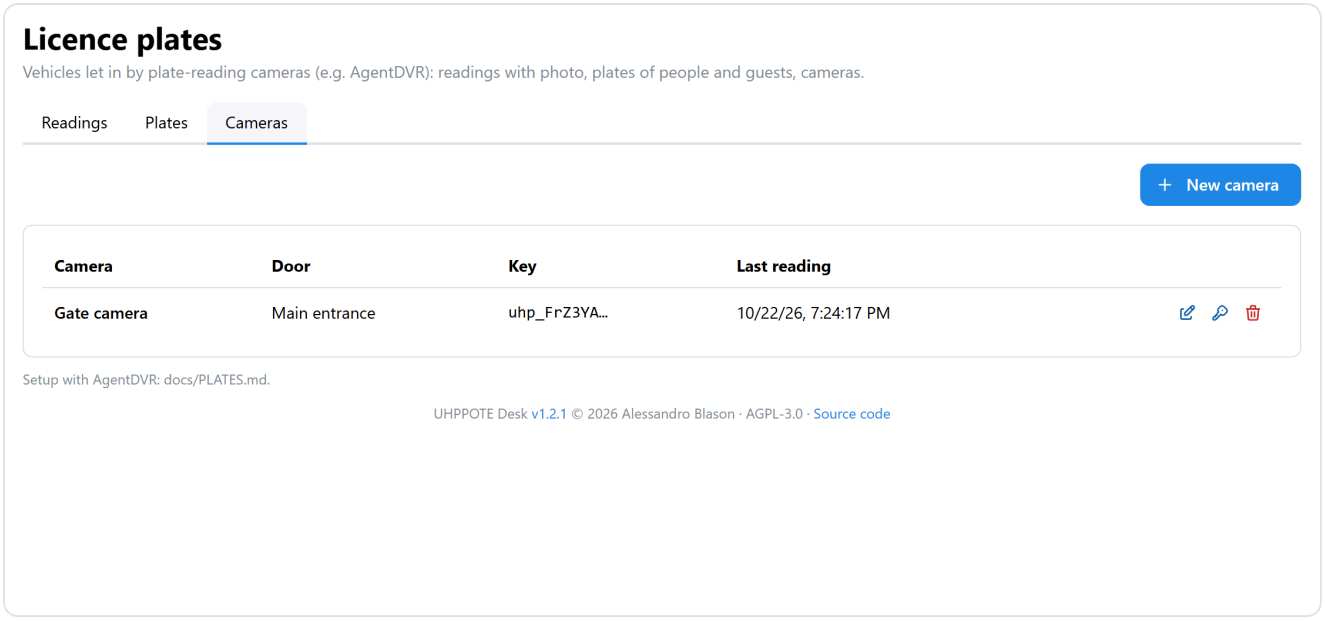



**Figure 13** — Plate-reading cameras.

When a camera is created, UHPPOTE Desk shows two addresses containing the camera's secret key, to copy into AgentDVR (list of known plates and "Call URL"): the key is shown only once. Readings are also published as `plate.granted` / `plate.denied` for notifications, rules and webhooks; Telegram and email notifications carry the photo. The step-by-step setup is in `docs/PLATES.md`.

PART 3

# Registry

People, cards and access permissions: who can go where, and when.

REGISTRY

## 12 · People

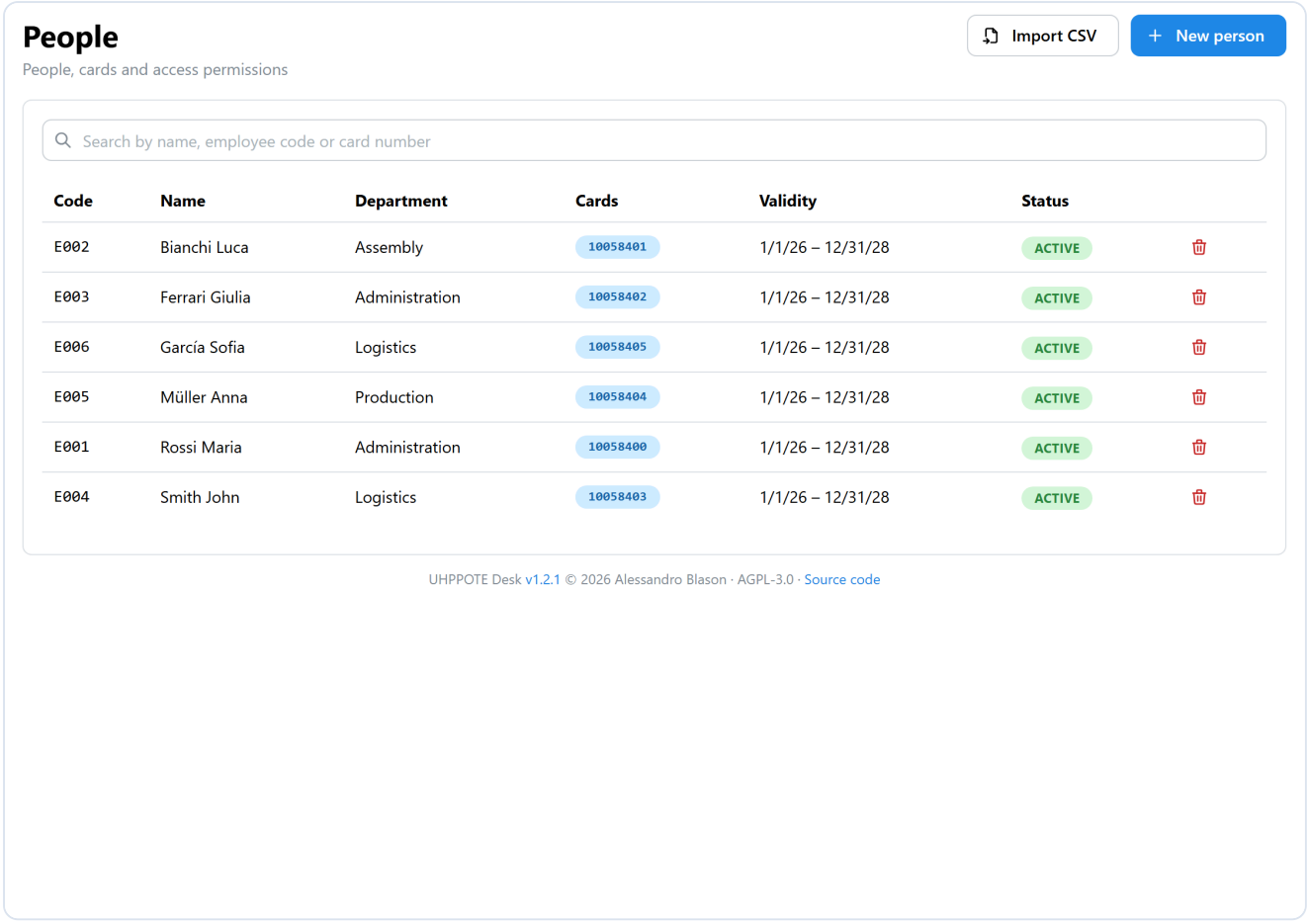



**Figure 14** — The list of people.

The list shows employee code, name, department, cards, validity and status. Search by name, employee code or card number. Click a row to open the details; New person creates one.

## Details

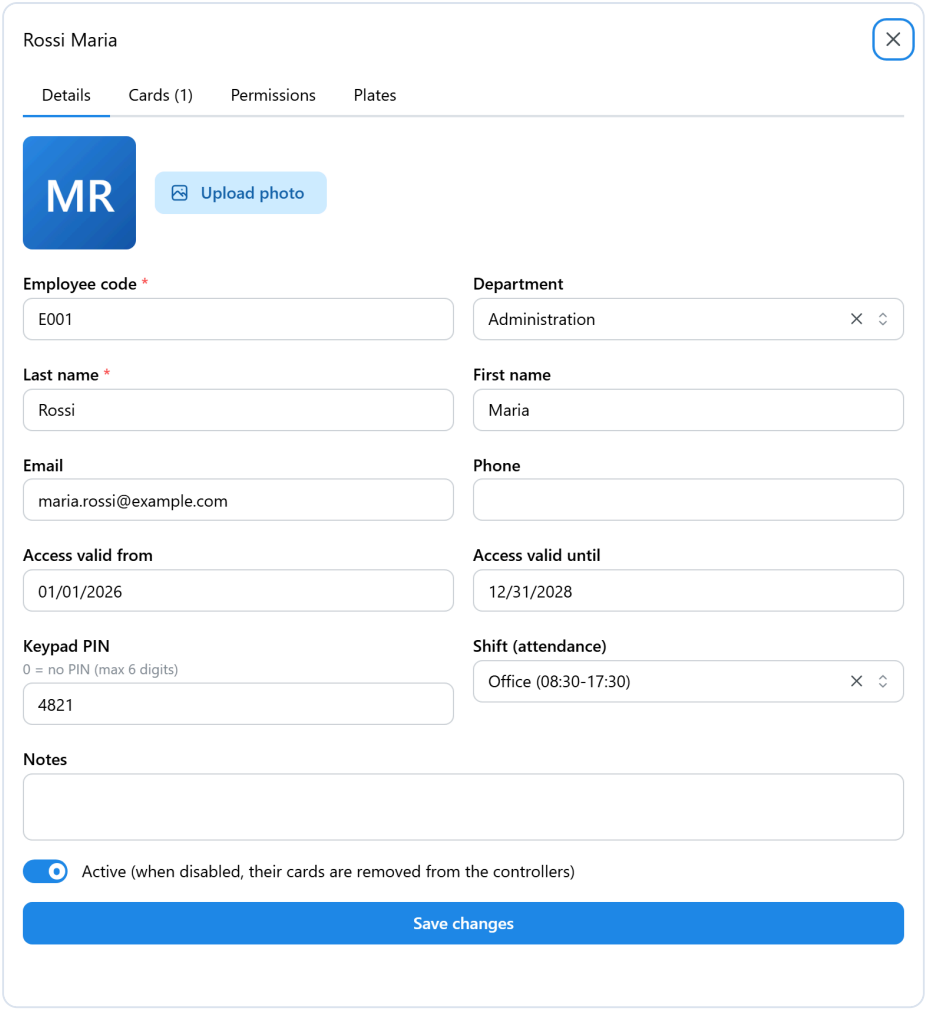



**Figure 15** — Person details.

- **Employee code** (unique) and **last/first name**; department, e-mail, phone, notes.
- **Access valid from / until** — outside this period all the person's cards are refused by the controllers.
- **Keypad PIN** (up to 6 digits, 0 = none) — requested by readers with a keypad when the door works in "card + PIN" mode.
- **Shift** — for attendance.
- **Photo** — JPEG, PNG or WebP; it is shown on the monitor.
- **Active** — when disabled, the person's cards are removed from all controllers (the history is kept). Use it for people who leave temporarily or permanently.

## Cards of a person

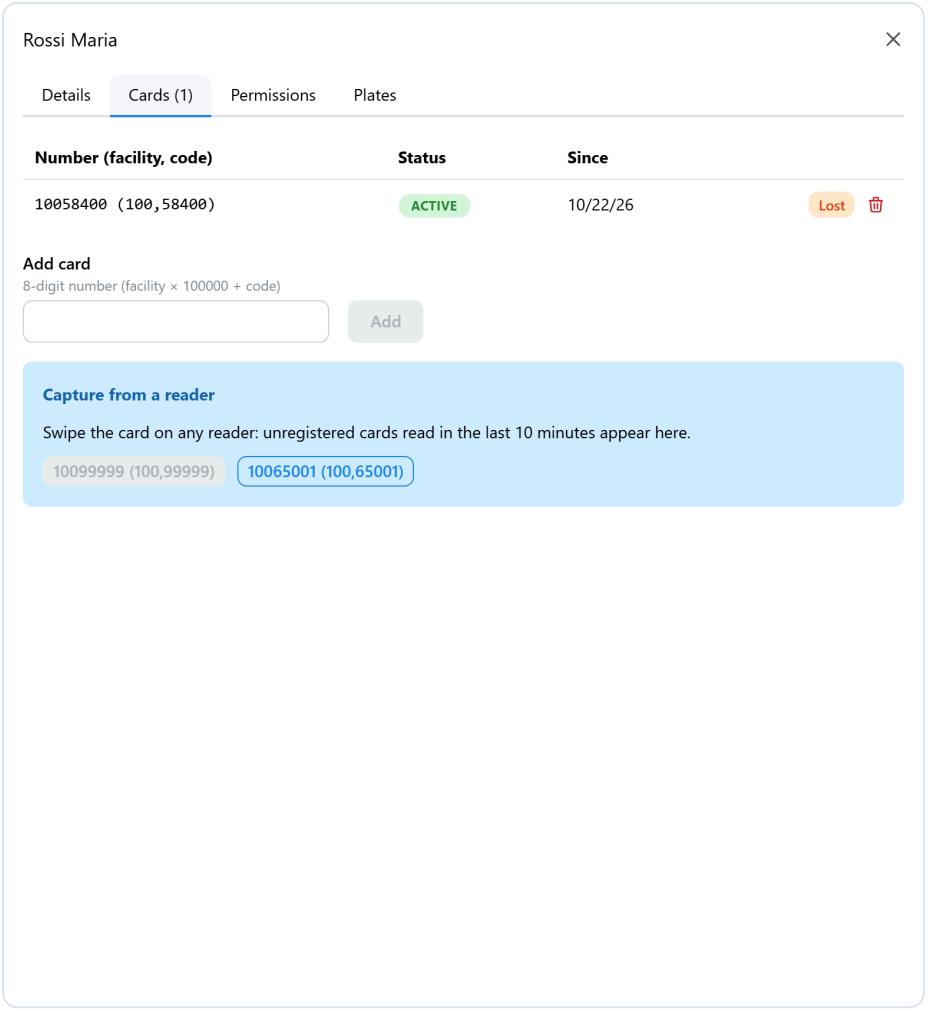



**Figure 16** — Cards tab.

Type the 8-digit card number and Add, or use **Capture from a reader**: swipe the new card on any reader and it appears among the unknown cards read in the last 10 minutes; click it to assign it. A person can have several cards.

- Lost — replaces a lost card: the old one is marked as lost and removed from the controllers, the new number gets the same permissions immediately.
- **Delete** — removes the card from the database and from every controller.

## Permissions of a person

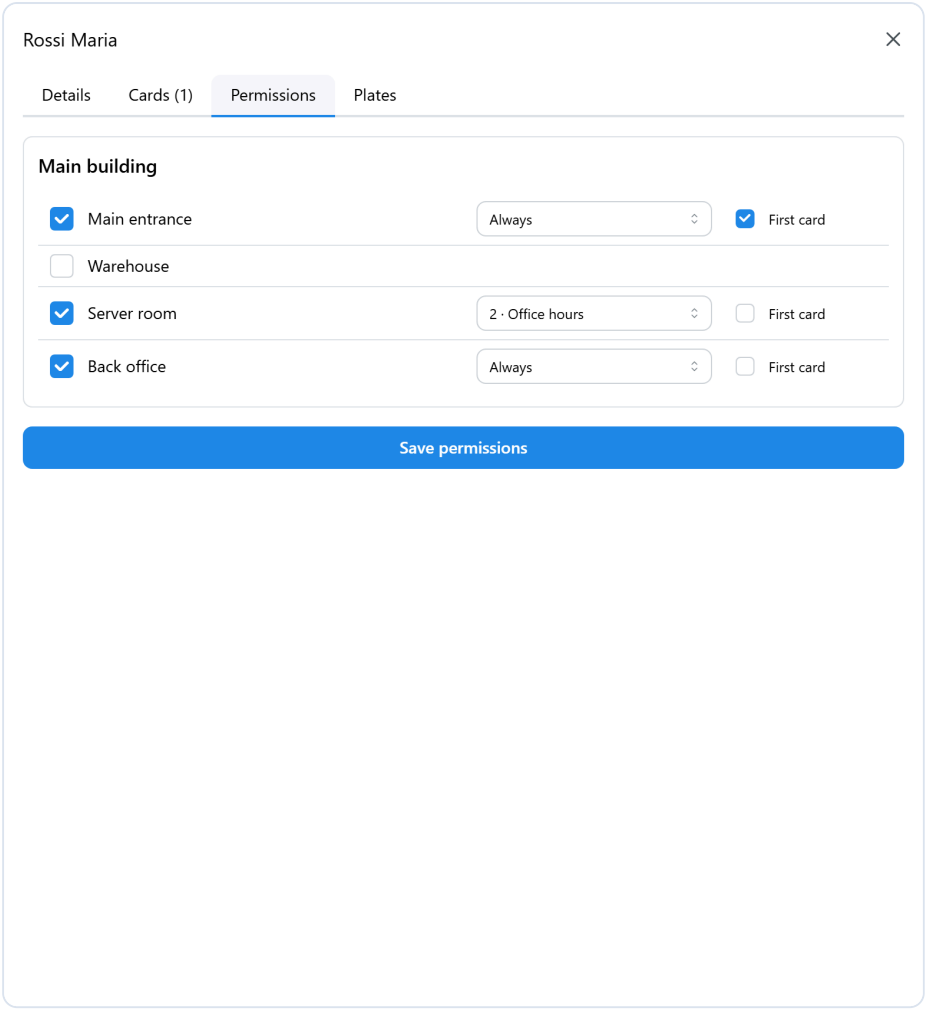



**Figure 17** — Permissions tab.

One row per door of every controller. Tick the doors the person may open and choose, for each, **Always** or a **time profile**. **First card** marks the person as able to "open" a door that uses the first-card function (see Controllers → Advanced settings). **Save permissions** sends them to the controllers immediately.

## Importing people from a CSV file

**Import CSV** reads a file with a header row; the separator (semicolon, comma or tab) is detected automatically. Recognised columns:

| Column | Accepted header names | Notes |
|---|---|---|
| Employee code | `code`, `user no`, `consumer no`, `matricola` | required; an existing code updates that person |
| Last name | `last name`, `lastname`, `name`, `cognome` | required |
| First name | `first name`, `firstname`, `nome` | |
| Card | `card`, `card id`, `card number`, `tessera` | 8-digit Wiegand-26 number |
| Department | `department`, `reparto` | created if missing; sub-departments with `\`, e.g. `Production\Assembly` |

The result shows how many people and cards were created or updated, and the rows with errors. Imported people have no permissions: assign them afterwards, for example with the bulk assignment of the Permissions page. An export of the original Windows software can usually be imported directly.

REGISTRY

## 13 · Cards

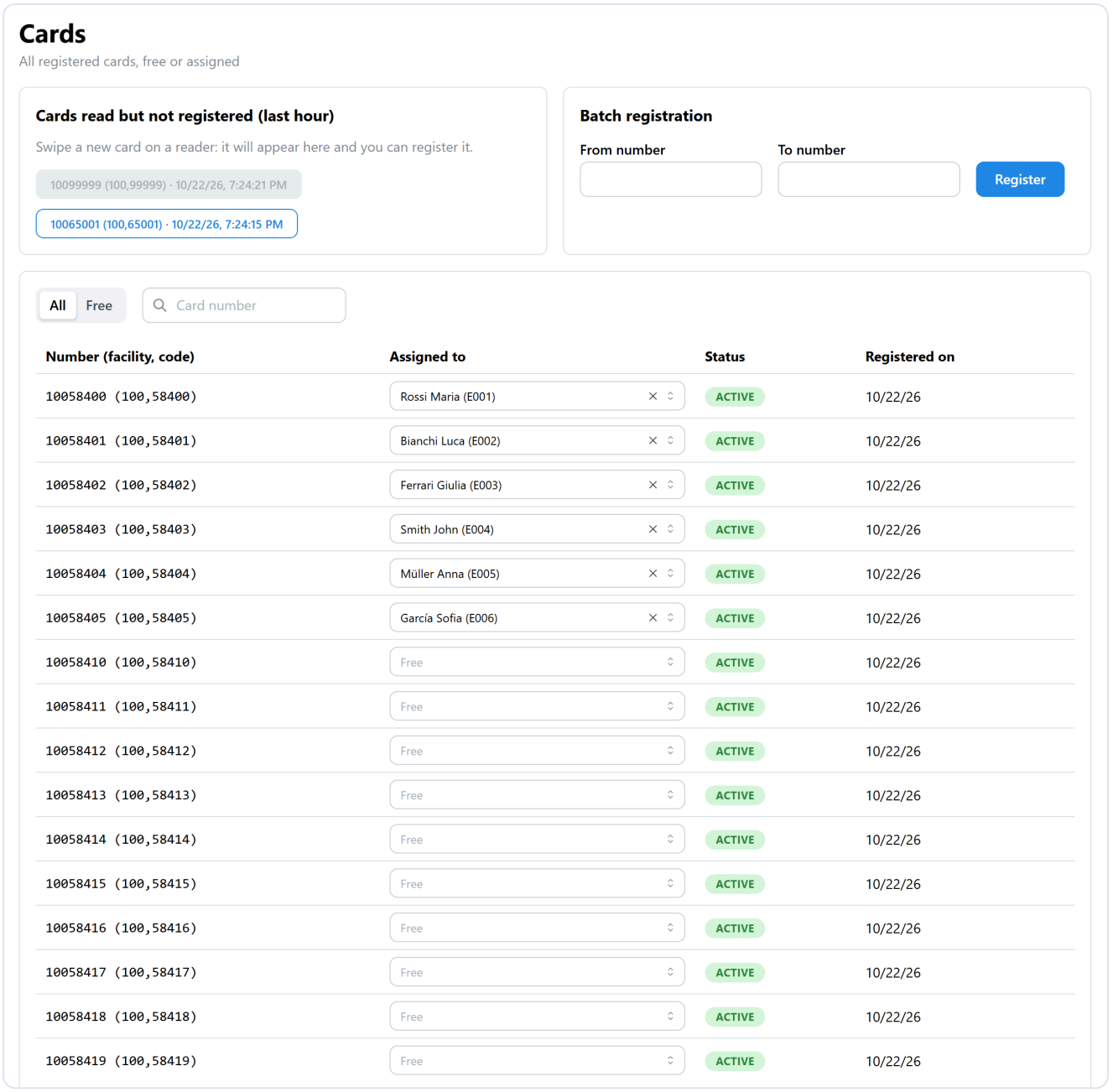



**Figure 18** — All registered cards.

All the cards known to the application, assigned or free, with status (Active, Lost, Disabled) and registration date. Filter **Free** to see the cards not yet given to anybody; a free card can be assigned from the list.

- **Cards read but not registered (last hour)** — every unknown card swiped on any reader appears here: click it to register it, as a free card or directly assigned to a person. Numbers that are not valid Wiegand-26 cards are shown greyed out and cannot be registered.
- **Batch registration** — registers a whole range of numbers as free cards (up to 1000), e.g. a new box of cards with consecutive numbers.

> **Card numbers**
> The controllers accept Wiegand-26 numbers only: facility code 0–255 and card code 0–65535, written as facility × 100000 + code. The number printed on many cards (e.g. `100,58400` or `0006612288`) corresponds to this format; if in doubt, swipe the card and read its number in the unknown cards list.

REGISTRY

# 14 · Permissions

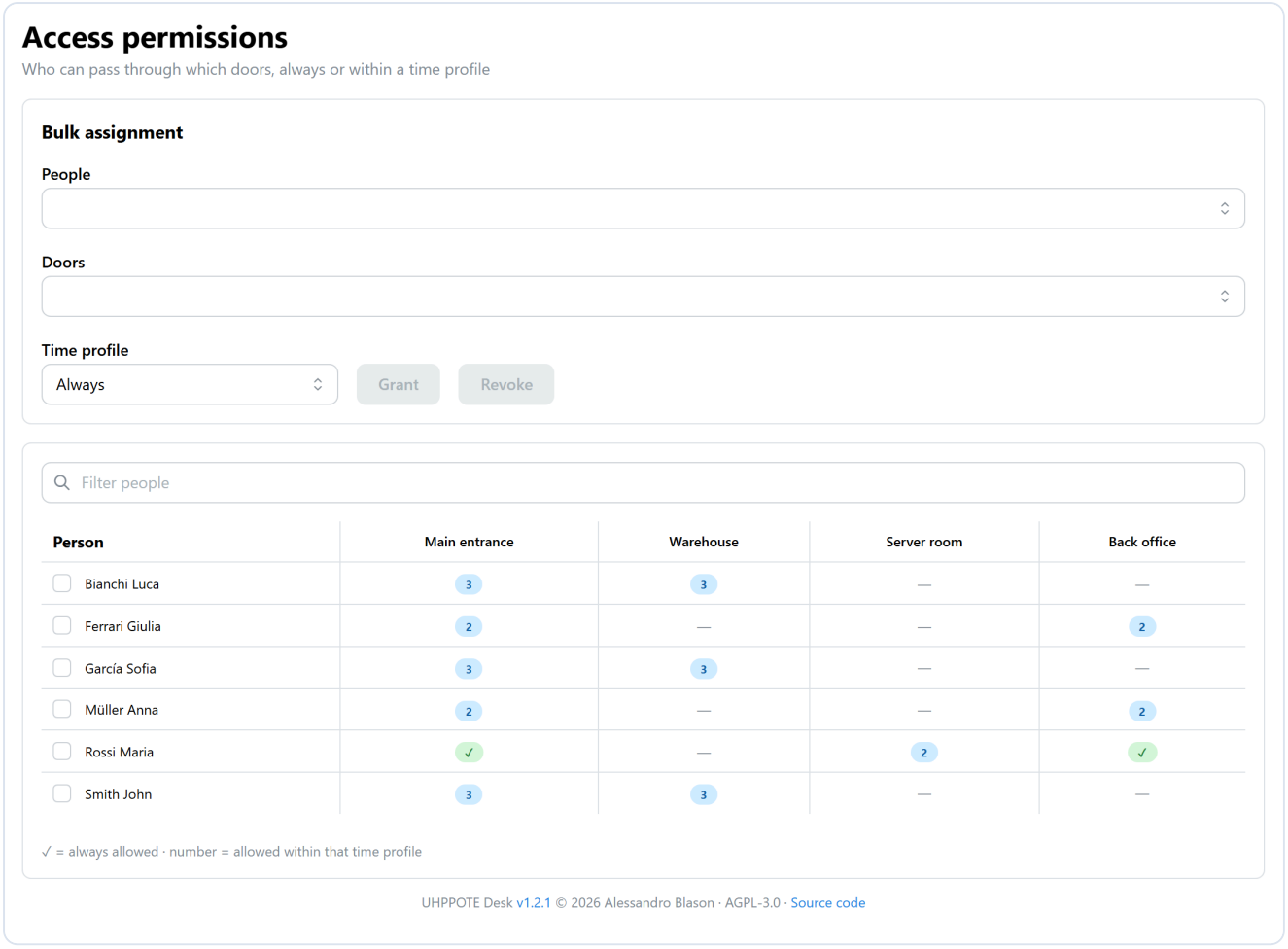



**Figure 19** — Matrix of people and doors, and bulk assignment.

The equivalent of "Access Privilege" in the original software: a matrix with one row per person and one column per door. ✓ means always allowed, a number means allowed within that time profile. Filter the people to find someone quickly.

With **Bulk assignment** you select several people and several doors, optionally a time profile, and Grant or Revoke all those permissions in one go — for example to give a whole new department access to the entrance.

REGISTRY

# 15 · Departments and zones

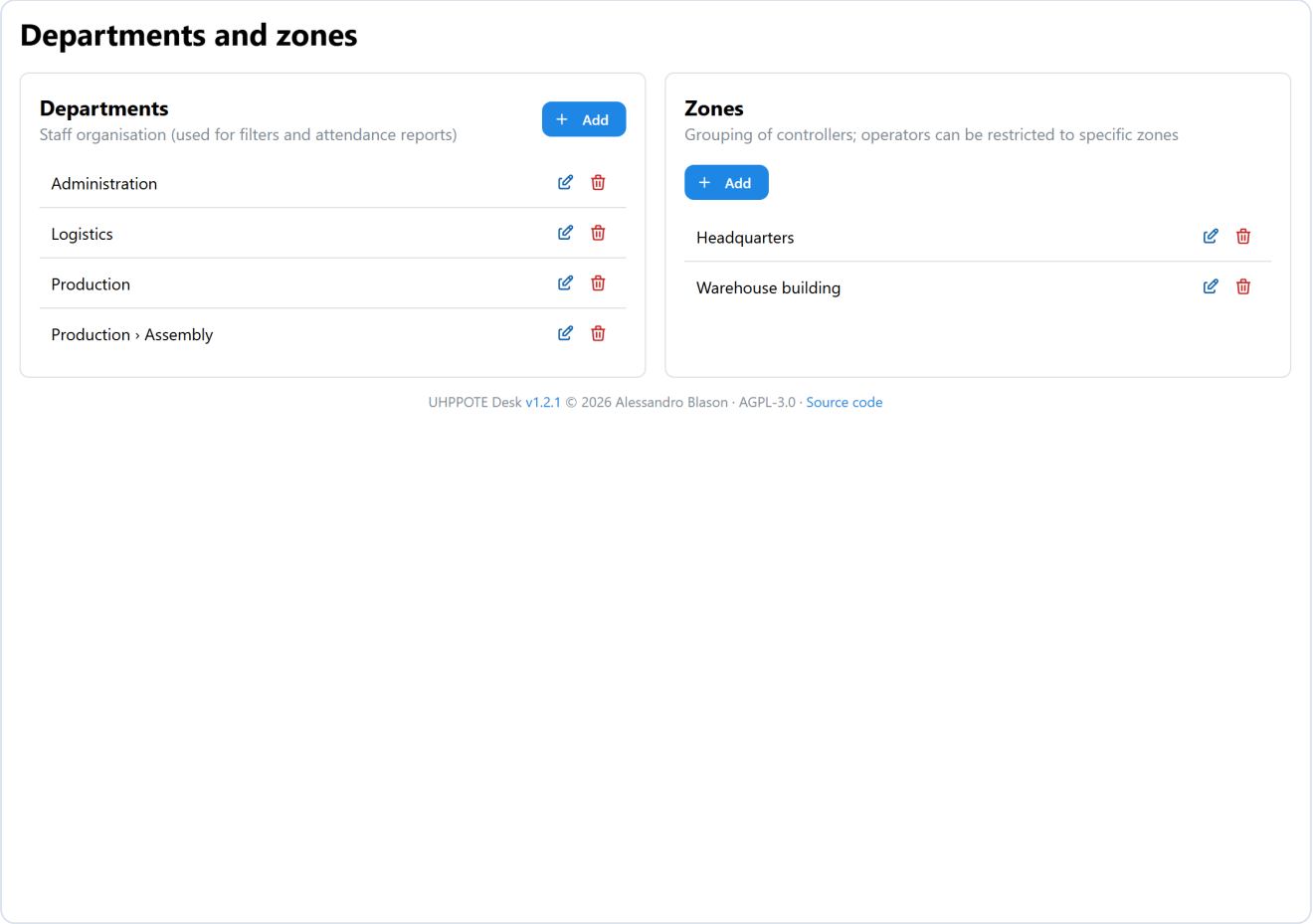



**Figure 20** — Departments and zones are trees.

- **Departments** organise the staff: they are used to filter the list of people and the attendance reports. A department can belong to another one (e.g. Production → Assembly).
- **Zones** group controllers, e.g. by building or floor. They are used to restrict operators: an operator restricted to a zone (and its sub-zones) can open doors and synchronise only the controllers of that zone.

Use New, edit and delete; deleting an item also deletes its sub-items. People and controllers of a deleted department or zone are kept, without department or zone.

REGISTRY

# 16 · Guests

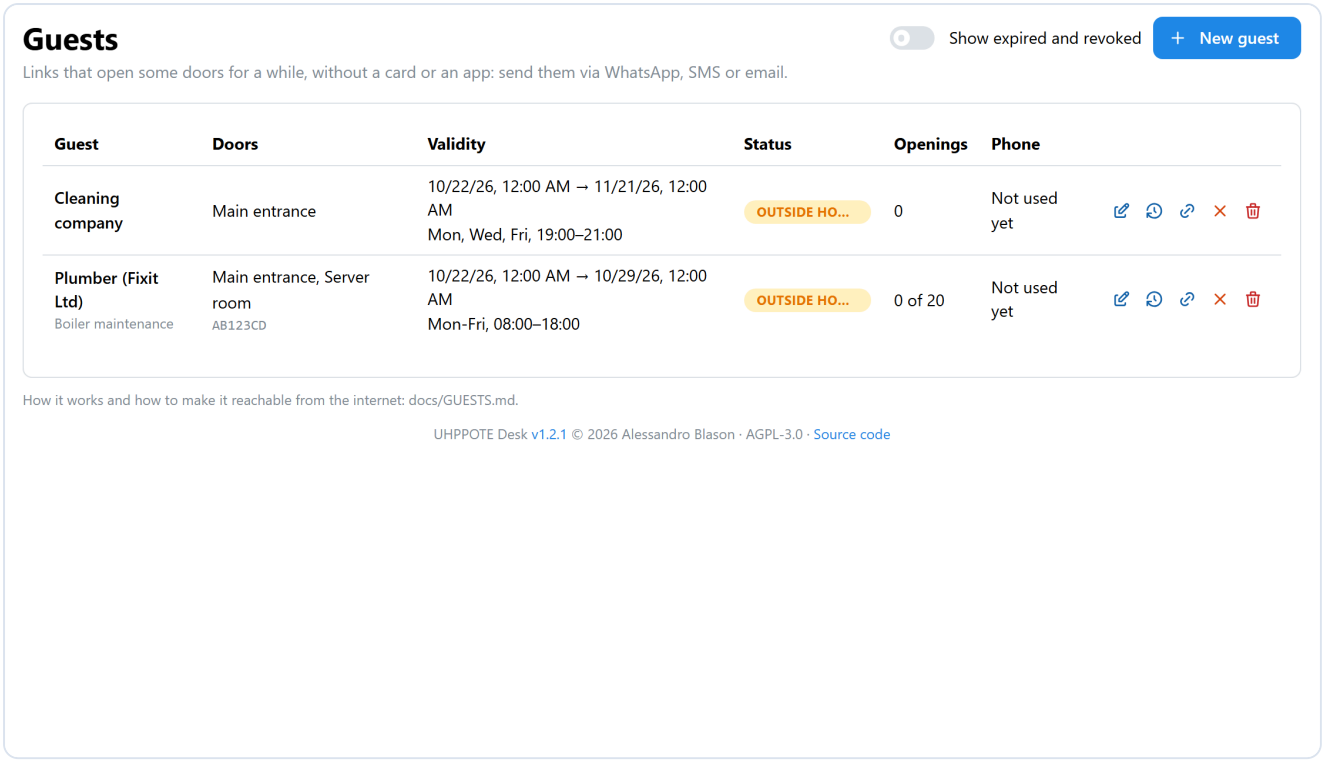



**Figure 21** — Guest links.

A guest link opens some doors for a while, **without card, app or login**: a courier or a technician today, a cleaning company every Tuesday evening, a friend for the weekend. Operators create links only for the doors they may open themselves; administrators see all of them.

- **Doors** and **validity** (from, until; buttons *Today*, *7 days*…), optionally **only on some days and hours** (e.g. Monday–Friday 08:00–18:00) and a **maximum number of openings**.
- **Only on the first phone that uses it** (on by default): a forwarded link does not work on another phone, and the attempt is recorded and notified.
- **PIN** (4–8 digits), to give separately from the link; after 5 wrong PINs the link is blocked until it is unlocked.
- **Vehicle plates**: plate-reading cameras open the doors of the link for these vehicles (see Licence plates).

After saving, the message to send is shown with Share, WhatsApp and copy buttons. **The link is shown only at that moment** (only its hash is stored): if it is lost, New link replaces it and the old one stops working. From the list you can also revoke a link, release it from the phone it is bound to, and see its openings.

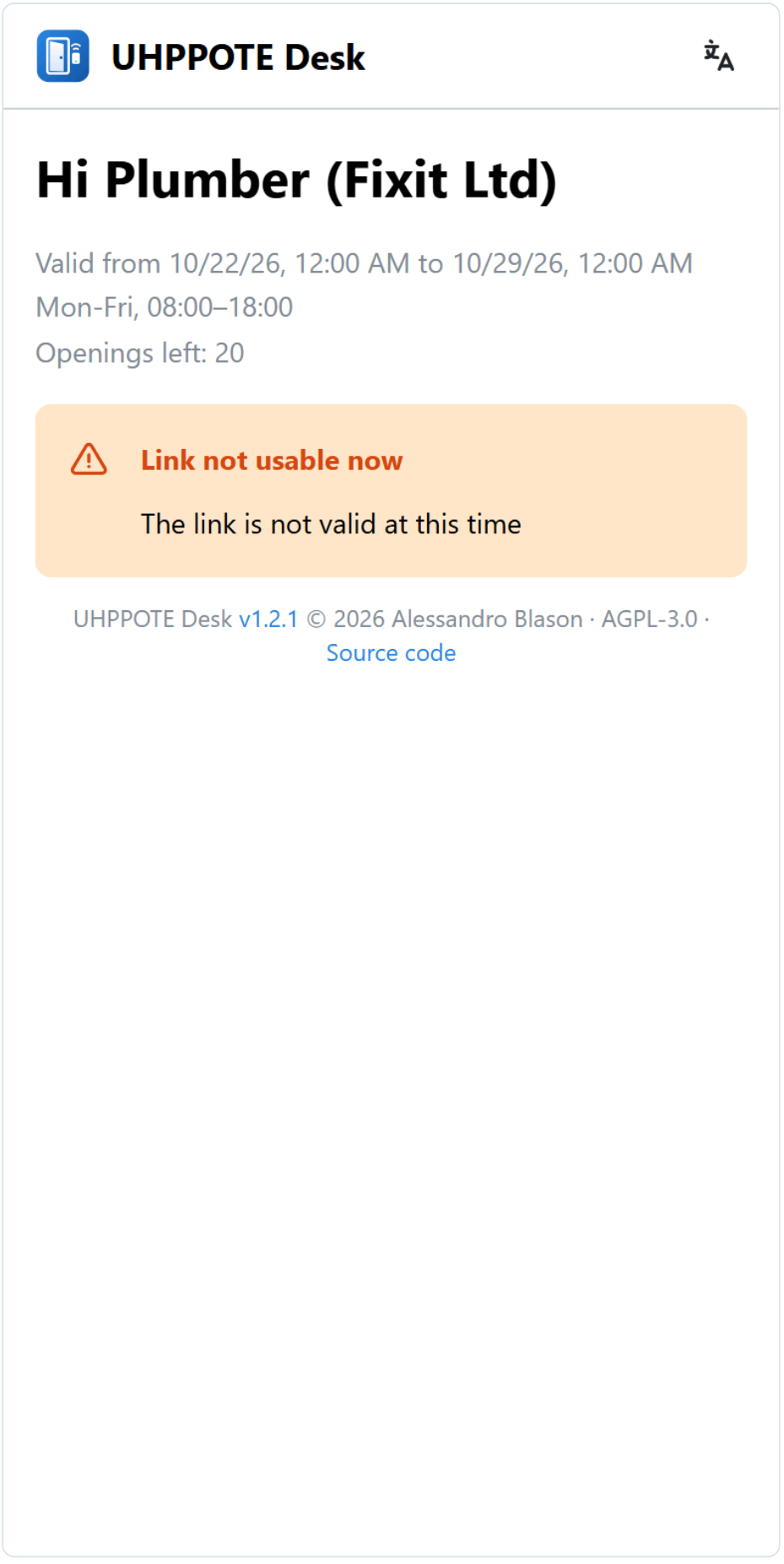



**Figure 22** — What the guest sees on the phone: one "Hold to open" button per door.

The guest page shows the name, the validity and one **Hold to open** button per door (a short touch does not open). When the link cannot be used it explains why: not valid yet, outside the hours, expired, used up,

revoked or blocked. Openings and refusals are in the history and are published as `guest.opened` / `guest.refused`. Details: `docs/GUESTS.md`.

PART 4

# Configuration

Controllers, doors and the rules they apply. Most of these pages are for administrators.

CONFIGURATION

## 17 · Controllers

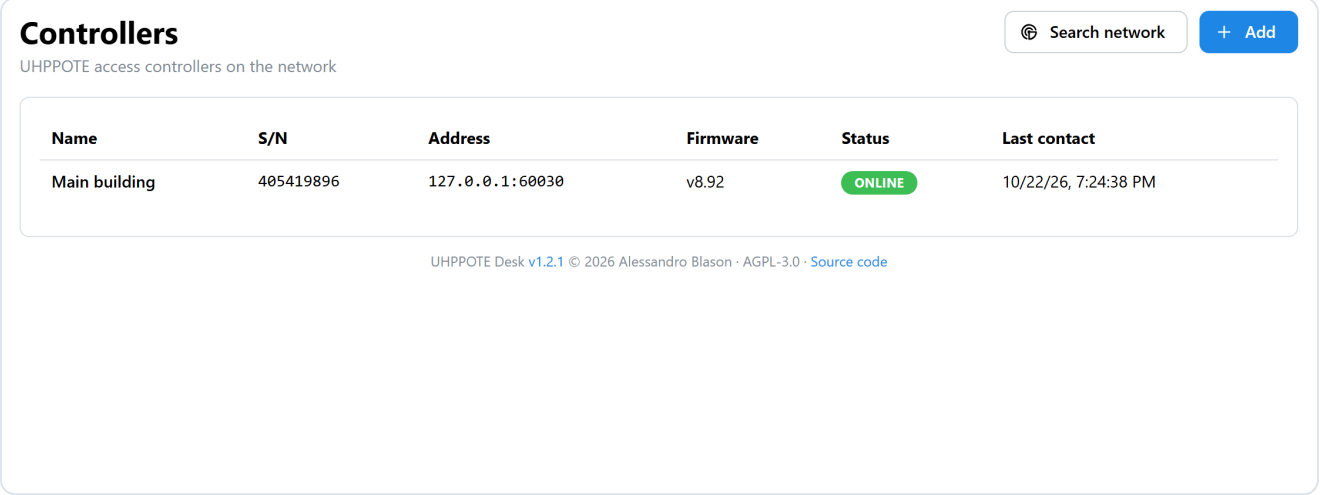



**Figure 23** — The controllers configured in the application.

The list of controllers with name, serial number, address, firmware, status and last contact. Click a row to open its page.

### Finding and adding a controller

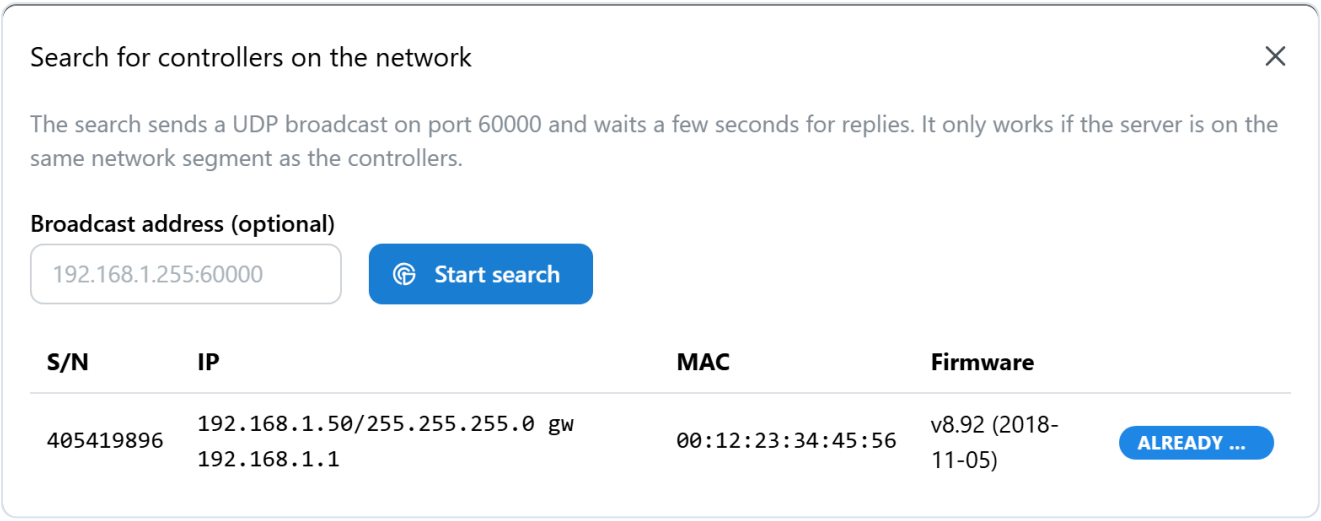



**Figure 24** — Network search.

1. Click Search network, then Start search. The server sends a broadcast on the local network and lists every controller that answers within a few seconds, with serial number, IP, MAC and firmware.

2. Click **Add** next to a controller. Give it a name, check IP address and port (60000), transport (UDP; TCP only for recent firmware) and optionally a zone.
3. Save: the controller appears online within a few seconds. The application programs it to send its events to the server, synchronises its clock and downloads its stored events.

A controller can also be added manually with its serial number. The search works only if the server is on the same network segment as the controllers; with Docker, the container must use the "host" network.

> **Controller already in use?**
> If the controller already contains cards (for example from the original software) and the application database has none, nothing is deleted when you add it. Import the people first, then use Synchronisation → Compare to see the differences before aligning.

## Controller page — Diagnostics

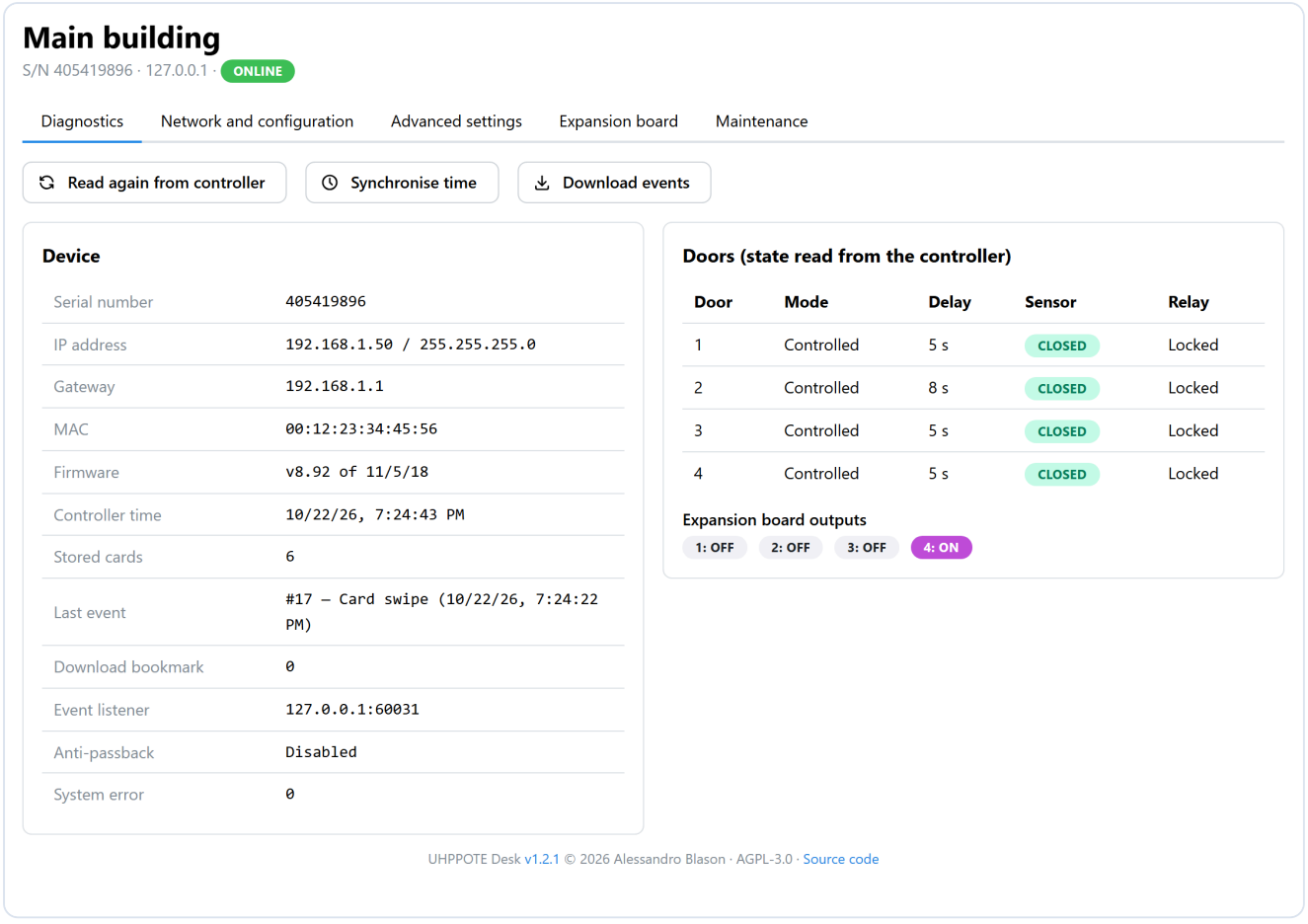



**Figure 25** — Diagnostics: everything read live from the controller.

Read directly from the controller: serial number, IP, gateway, MAC, firmware version and date, controller time, number of stored cards, last event, download bookmark, event listener (where the controller sends its events), anti-passback and system error. The **Doors** table shows mode, opening delay, sensor and relay of each door, as stored in the controller.

- **Read again from controller** refreshes the data.
- **Synchronise time** sets the controller clock to the server time. This also happens automatically when the drift exceeds 30 seconds.
- **Download events** downloads at once the events not yet collected.

## Network and configuration

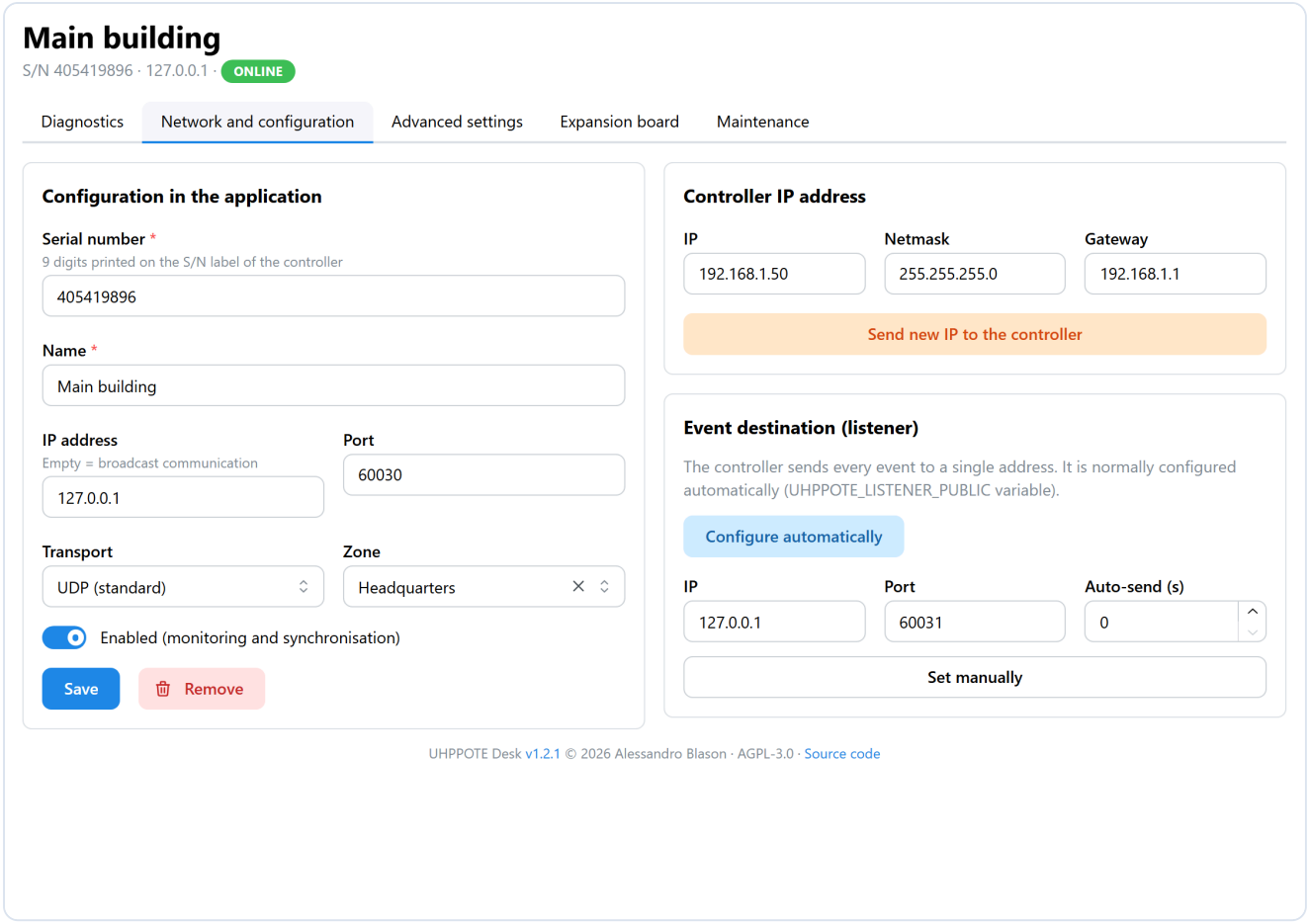



**Figure 26** — Configuration in the application, controller IP and event destination.

- **Configuration in the application** — how the application reaches the controller: serial number, name, IP and port (empty IP = broadcast), transport, zone and **Enabled** (a disabled controller is neither monitored nor synchronised). Remove deletes the controller and its event history from the database, without changing the controller.
- **Controller IP address** — changes the IP, netmask and gateway stored **in the controller**. Use it to move a new controller from its factory address to your network. The command is sent by broadcast, so it works even if the current address is not reachable; afterwards the application checks the new address.
- **Event destination (listener)** — the address to which the controller sends every event: the server. It is set automatically when the server is configured with `UHPPOTE_LISTENER_PUBLIC`; Configure automatically applies it now, Set manually sets a different address and the status auto-send interval.

> **Careful**
>
> A wrong IP address, netmask or gateway makes the controller unreachable. It can then be reconfigured only from a computer on the same network segment, with a network search.

# Advanced settings

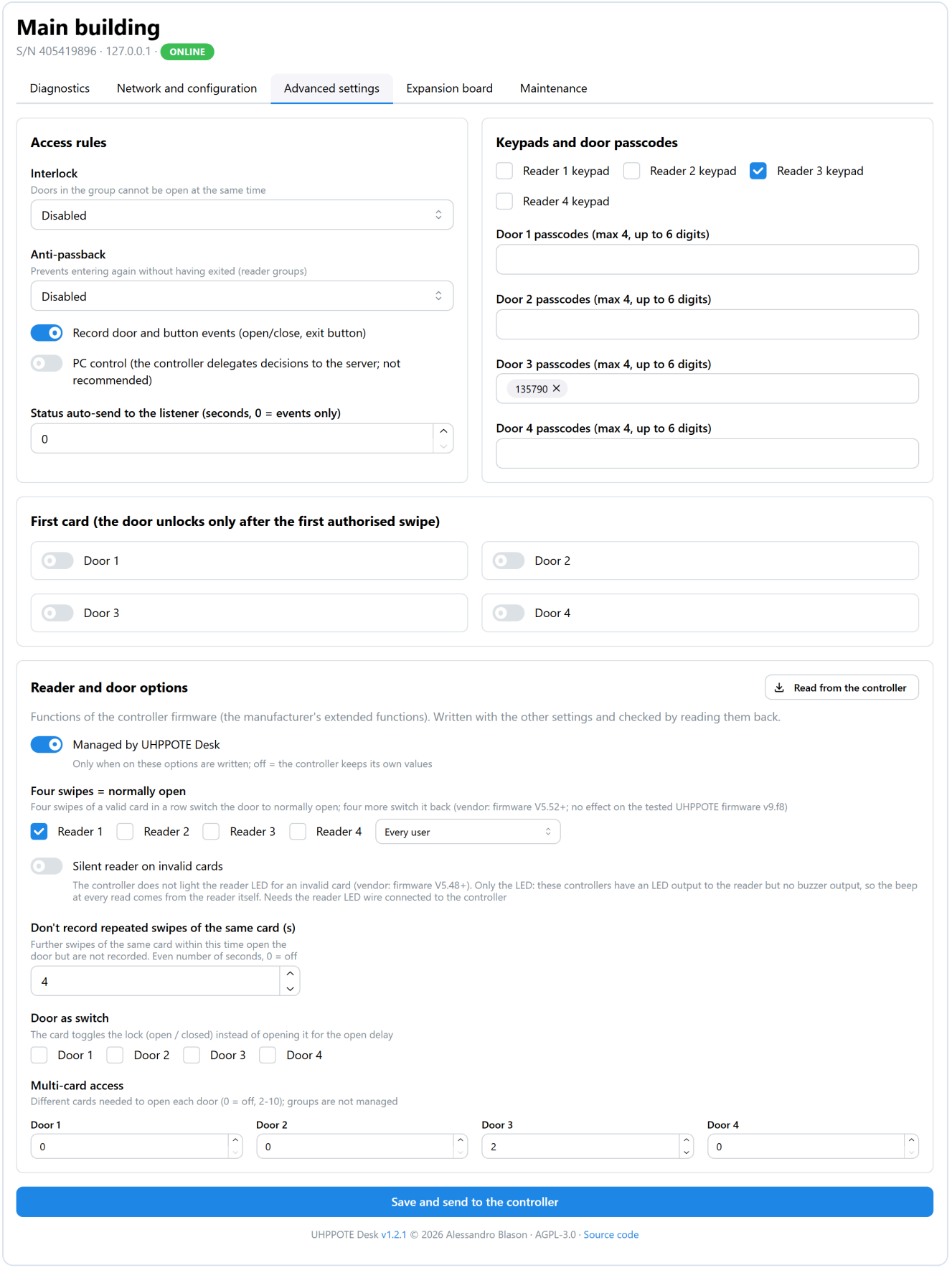



**Figure 27** — Access rules, keypads and first card.

Settings that apply to the whole controller. They are stored in the database and sent to the controller with **Save and send to the controller**.

- **Interlock** — doors of a group cannot be open at the same time (airlocks, e.g. doors 1 and 2): a door opens only when the others of its group are closed. Requires door sensors.
- **Anti-passback** — prevents passing a card back to someone else: after entering through a reader of one group, the card must exit through a reader of the other group before it can enter again. See chapter 34 for the groupings.
- **Record door and button events** — also records door opened/closed and exit button events (recommended).
- **PC control** — the controller asks the server before granting access. Not recommended: doors stop working when the server is unreachable.
- **Status auto-send** — the controller sends its status to the server every N seconds besides the events (0 = events only).
- **Keypads** — tick the readers that have a keypad. **Door passcodes**: up to 4 codes per door (up to 6 digits) that open the door when typed on the keypad, without a card.
- **First card** — in the chosen hours and days the door stays locked until the first authorised person with the "First card" permission swipes in; then the door switches to the chosen mode (e.g. normally open) until the end of the period. Typical use: a shop entrance that opens to the public only after the manager has arrived.
- **Reader and door options** — the corresponding *extended functions* of the manufacturer's software, written only when Managed by UHPPOTE Desk is on and checked by reading them back: **four swipes = normally open** (four swipes of a valid card in a row keep the door open, four more lock it again; stored but without effect on the tested firmware v9.f8), **silent reader** on invalid cards (the controller does not light the reader LED; these controllers have no buzzer output, so the beep the reader makes at every read stays), **don't record repeated swipes** of the same card (further swipes within the interval open the door but are not recorded again), **door as switch** (a swipe toggles the lock) and **multi-card access** (2–10 different cards needed to open). Read from the controller shows the values the controller stores.

# Expansion board

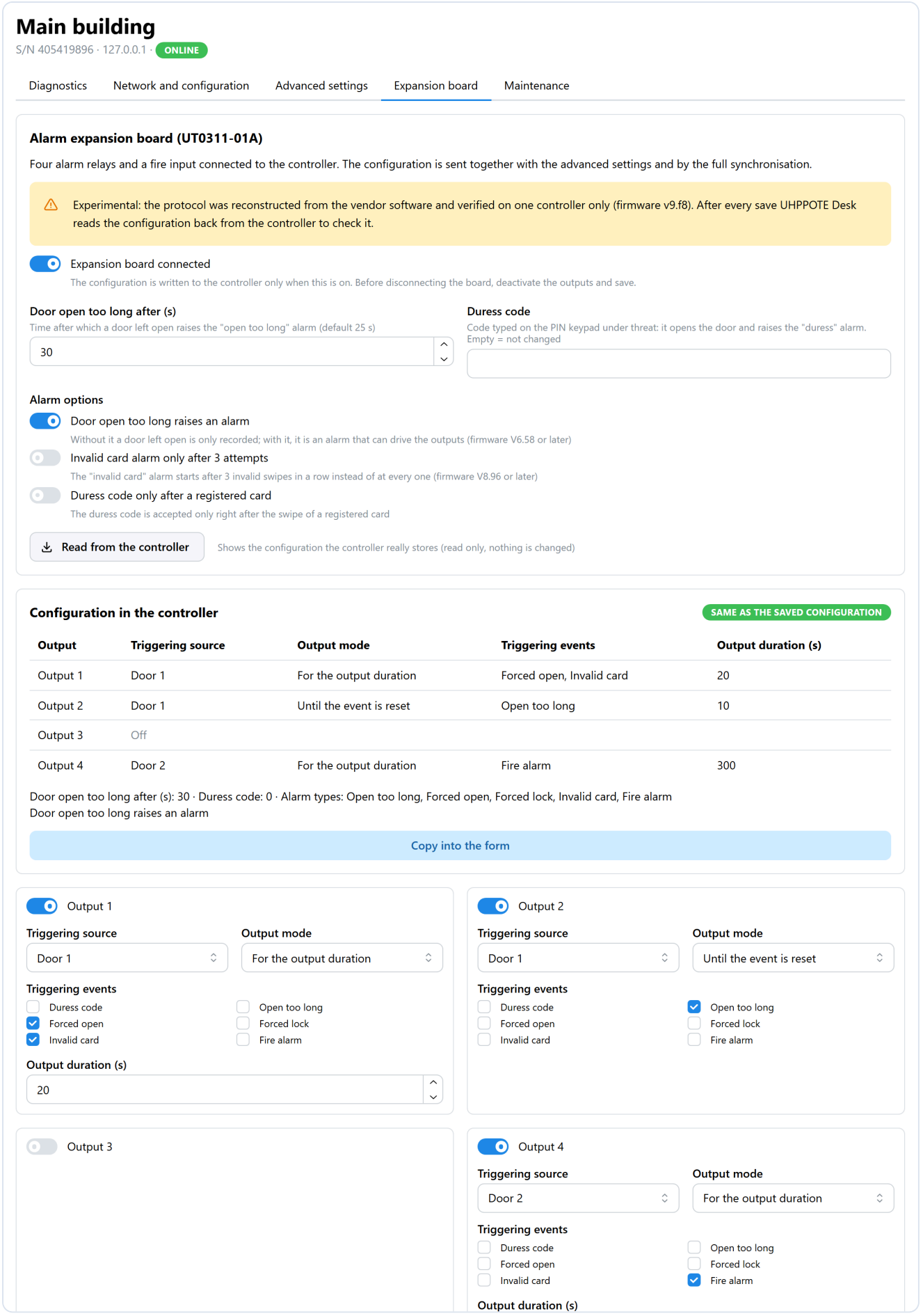



**Figure 28** — Alarm expansion board: four outputs and the fire input.

The UHPPOTE **UT0311-01A** alarm expansion board (also sold as ACM-EC05) adds four alarm relays and a fire input to the controller: a siren, a warning light or a signal to a burglar or fire panel can be switched on when something happens at the doors. The configuration is in the **Expansion board** tab; **Save and send to the controller** writes it and then reads it back from the controller to check it (the message says *verified* when the controller stores exactly what was sent); the full synchronisation writes it again.

- **Expansion board connected** — the configuration is written to the controller only when this is on, so controllers without the board are never touched.
- **Door open too long after** — seconds after which a door left open raises the *Door open too long* alarm (default 25 s).
- **Duress code** — a code typed on the PIN keypad under threat: the door opens as usual and the *duress* alarm is raised. Leave it empty to keep the code stored in the controller.
- For each **output**: the **triggering source** (one of the four doors, or the burglar alarm), the **output mode**, the **triggering events** and the **output duration**.
- **Alarm options** — **door open too long raises an alarm** (otherwise a door left open is only recorded; needed for the outputs to react to it), **invalid card alarm only after 3 attempts**, and **duress code only after a registered card**. They are the matching options of the manufacturer's *extended functions*.
- **Read from the controller** — shows the configuration the controller really stores (read only, nothing is changed), with a badge saying whether it is the same as the saved one. **Copy into the form** takes it as a starting point, for example a configuration made with the manufacturer's software; save to apply it.

| Output mode | The relay stays on… |
|---|---|
| For the output duration | for the set number of seconds |
| For the duration and until reset | at least for the set time, and until the cause of the alarm is gone |
| Until the event is reset | until the cause of the alarm is gone (e.g. the door is closed again) |
| Any signal / All signals | while at least one (or all) of the selected signals is active: lock relay, door sensor, exit button, force lock input, inputs A1–A3 and fire input of the board |

Triggering events: **duress code**, **door open too long**, **forced open** (door opened without a valid access; requires a door sensor), **forced lock**, **invalid card** and **fire alarm**. The fire alarm also opens all the doors of the controller and is recorded as event 39.

**Live state.** For a controller with *Expansion board connected*, the **Diagnostics** tab shows the four outputs (*On/Off*, under the door table) and the dashboard card shows which ones are on. They are read from the same status the manufacturer's console uses, at every status check and, after any event or while something is on, every 2 seconds until everything is off. The fire alarm is shown at once and cleared a few seconds after the fire input is released.

> **Reconstructed protocol**
> The commands for the expansion board are not part of the public controller protocol: they were reconstructed from the manufacturer's software. They have been verified on a UT0311-L04 (firmware v9.f8) with a UT0311-01A board connected: forced open, door open too long, invalid card, fire input and outputs following a signal switch the relays, and the live state of the outputs is read correctly. Not verified yet: the duress code, the **burglar alarm** source and the inputs A1–A3. A configuration made with the manufacturer's software is replaced at the next save: read it and copy it into the form first if you want to keep it. Before removing the board, deactivate the outputs and save.

## Maintenance

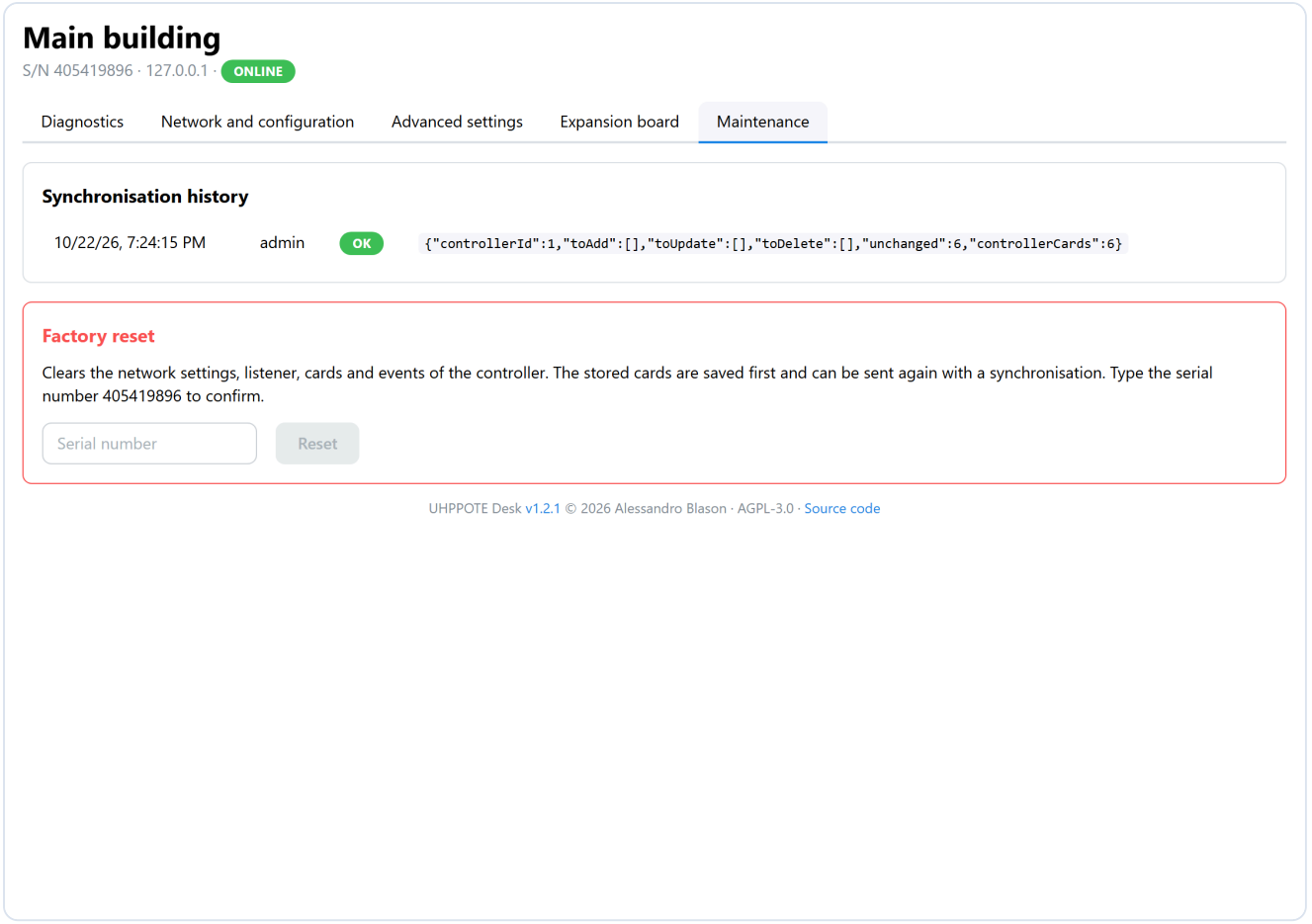



**Figure 29** — Synchronisation history and factory reset.

**Synchronisation history** lists the synchronisations of this controller, with operator, outcome and summary. **Factory reset** clears the network settings, listener, cards and events of the controller. The stored cards are first saved to a file in the server backup folder; to confirm, type the serial number. After a reset the controller must be configured again (IP, then synchronisation).

CONFIGURATION

# 18 · Doors

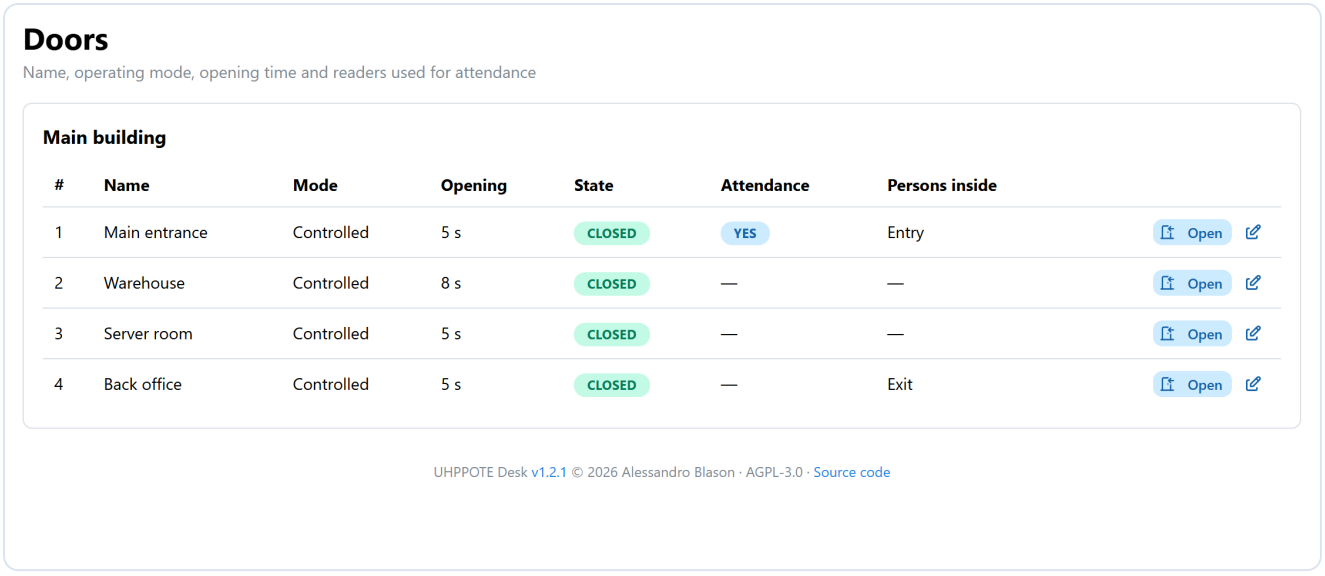



**Figure 30** — The doors of all the controllers.

Every door of every controller with its name, mode, opening time, live state and attendance flag. **Open** opens it remotely. Edit a door to change:

- **Name** — used everywhere in the application (dashboard, events, permissions).
- **Mode** — *Controlled* (normal: opens with an authorised card), *Normally open* (always unlocked) or *Normally closed* (always locked, even for valid cards).
- **Opening time** — seconds the lock stays released after a granted swipe, the exit button or a remote opening.
- **Active** — inactive doors (e.g. unused outputs) are hidden from the dashboard and from permissions.
- **Reader used for attendance** — swipes on this reader become attendance punches.
- **Persons inside** — *Entry*, *Exit* or *Not counted*: how the swipes on this door count on the Persons inside page.

Mode and opening time are sent to the controller immediately.

CONFIGURATION

# 19 · Time profiles

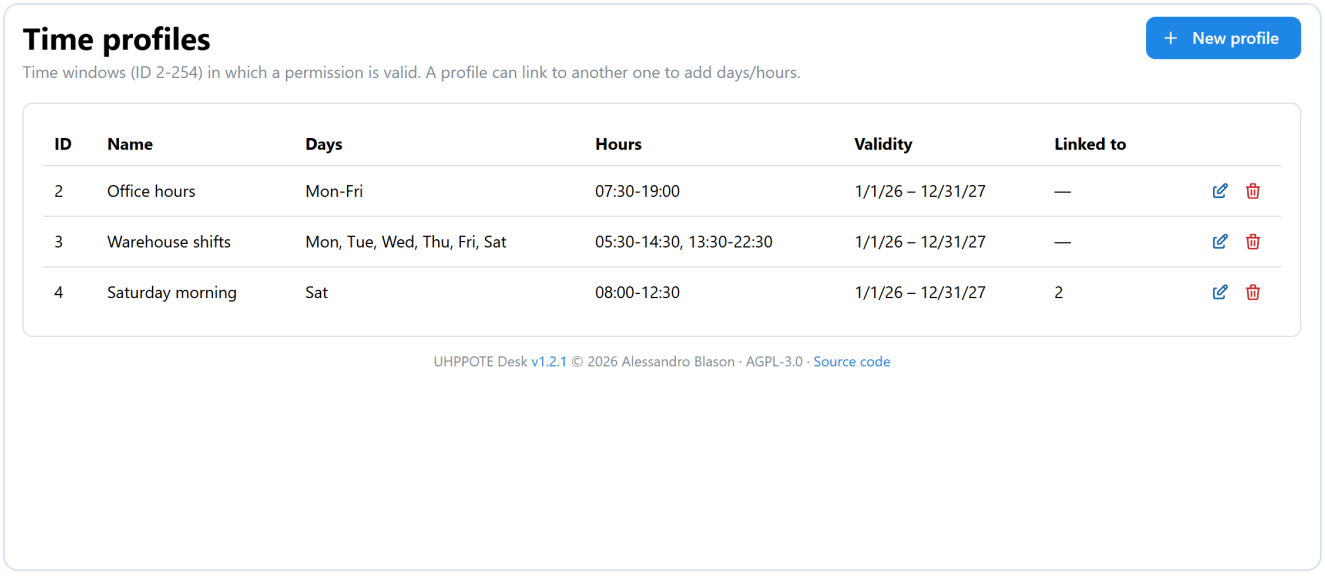



**Figure 31** — Time profiles.

A time profile tells **when** a permission is valid. Each profile has:

- an **ID** from 2 to 254 (the number stored in the controller) and a name;
- a validity period (**Valid from / until**);
- the **days** of the week;
- up to three **time intervals**, e.g. 08:00–12:30 and 13:30–18:00;
- optionally a **linked profile**: if a swipe does not fall within this profile, the linked one is checked too. Linking is the way to combine different hours on different days.

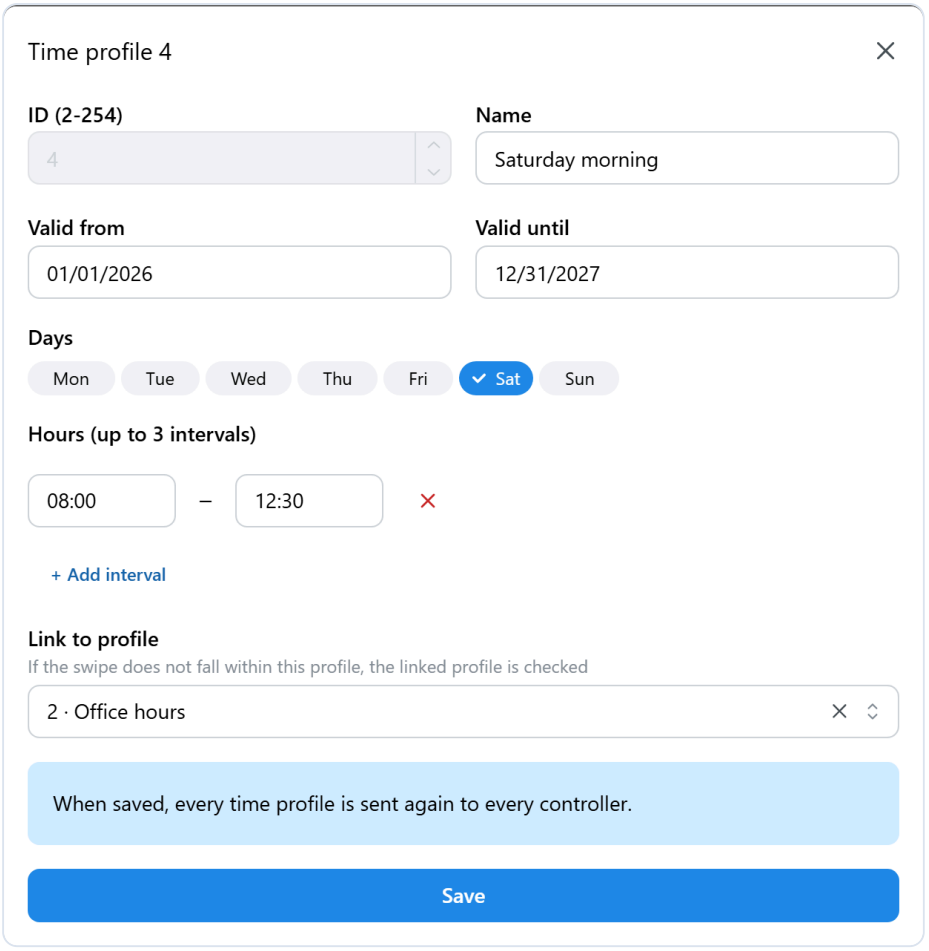



**Figure 32** — Editing a profile: "Saturday morning" is linked to "Office hours".

When a profile is saved, all the profiles are sent again to every controller. A profile used by some permission cannot be deleted.

> **Example**
> "Office hours" (2) = Mon–Fri 07:30–19:00; "Saturday morning" (4) = Sat 08:00–12:30, linked to 2. A permission with profile 4 allows access on weekdays during office hours *and* on Saturday morning.

CONFIGURATION

# 20 · Scheduled tasks

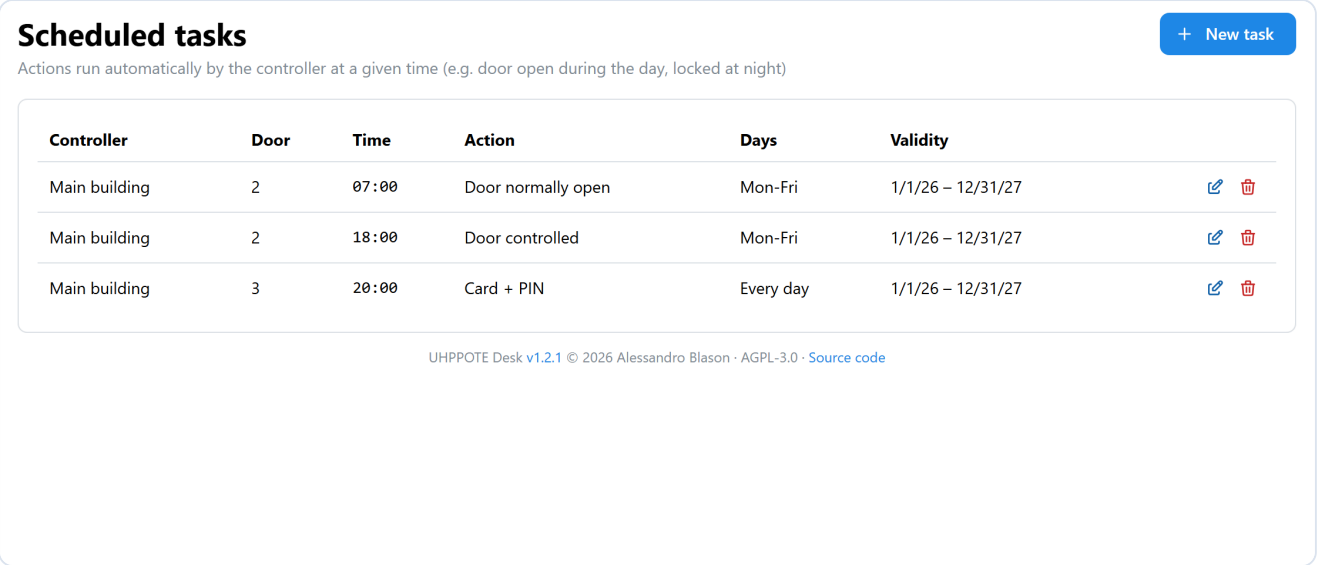



**Figure 33** — Scheduled tasks run by the controllers.

Actions that a controller runs **by itself** at a given time, on the chosen days and within a validity period — for example the warehouse door *normally open* at 07:00 and back to *controlled* at 18:00. Each task has controller, door, action, start time, days, validity and an Active switch. Available actions:

| Action | Effect from the start time |
|---|---|
| Door controlled / normally open / normally closed | changes the mode of the door |
| Disable / enable time profiles | time profiles are ignored (cards valid at any time) or applied again |
| Card only (no PIN) · Card + PIN on entry · Card + PIN | how readers with a keypad ask for the PIN |
| Enable / disable multi-card | access requires several different authorised cards in sequence (number of cards set in the task) |
| Single opening | opens the door once |
| Disable / enable exit button | ignores or re-enables the exit button |

CONFIGURATION

# 21 · Synchronisation

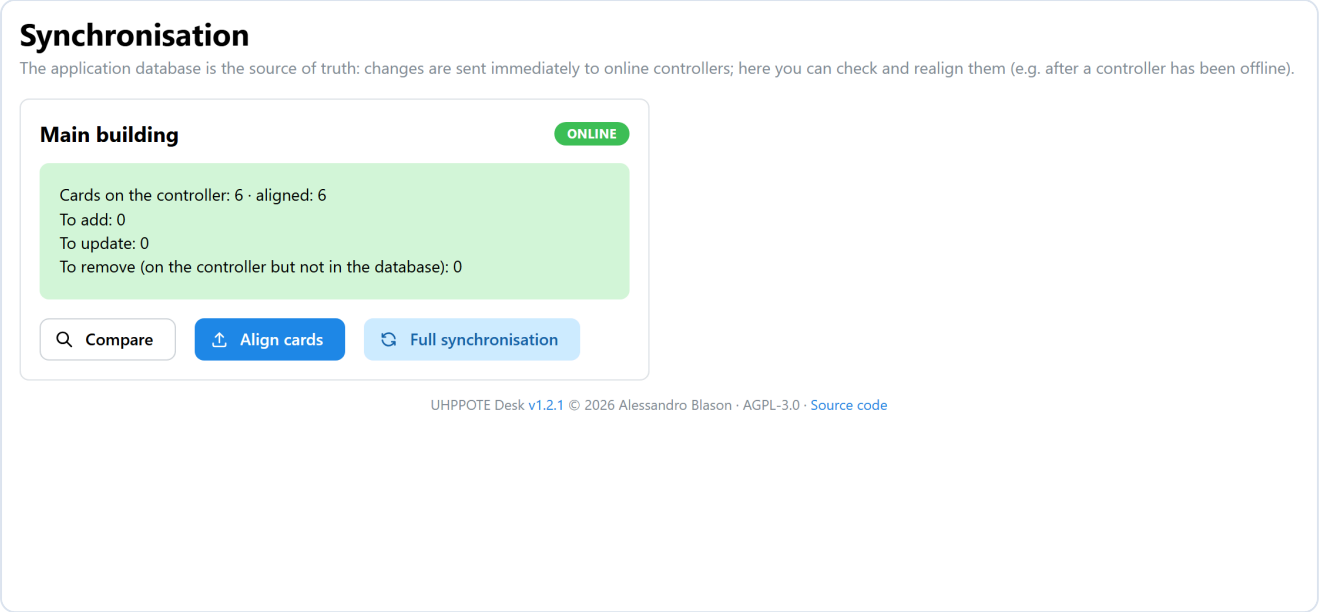



**Figure 34** — Comparing the database with a controller.

Every change is sent to the controllers when it is made. If a controller was unreachable at that moment (switched off, network fault), it is aligned here. The page lists the controllers with their state and these commands:

- **Compare** — reads all the cards from the controller and shows the differences: cards to add, cards to update (permissions, validity, PIN) and cards to remove (present on the controller but not in the database). Nothing is changed.
- **Align cards** — applies those differences to the cards.
- **Full synchronisation** (administrators) — sends again door configuration, time profiles, tasks, advanced settings and all the cards. It may take a few minutes with many cards; progress is shown live.

### Automatic recovery

When a controller comes back online **without any card** while the database expects some (factory reset, replaced controller), the application runs a full synchronisation by itself. Any other difference is only reported in the server log, so that cards are never deleted without an operator deciding it.

PART 5

# Administration

Users of the application, traceability and server maintenance. Administrators only, except the System page.

ADMINISTRATION

## 22 · Operators

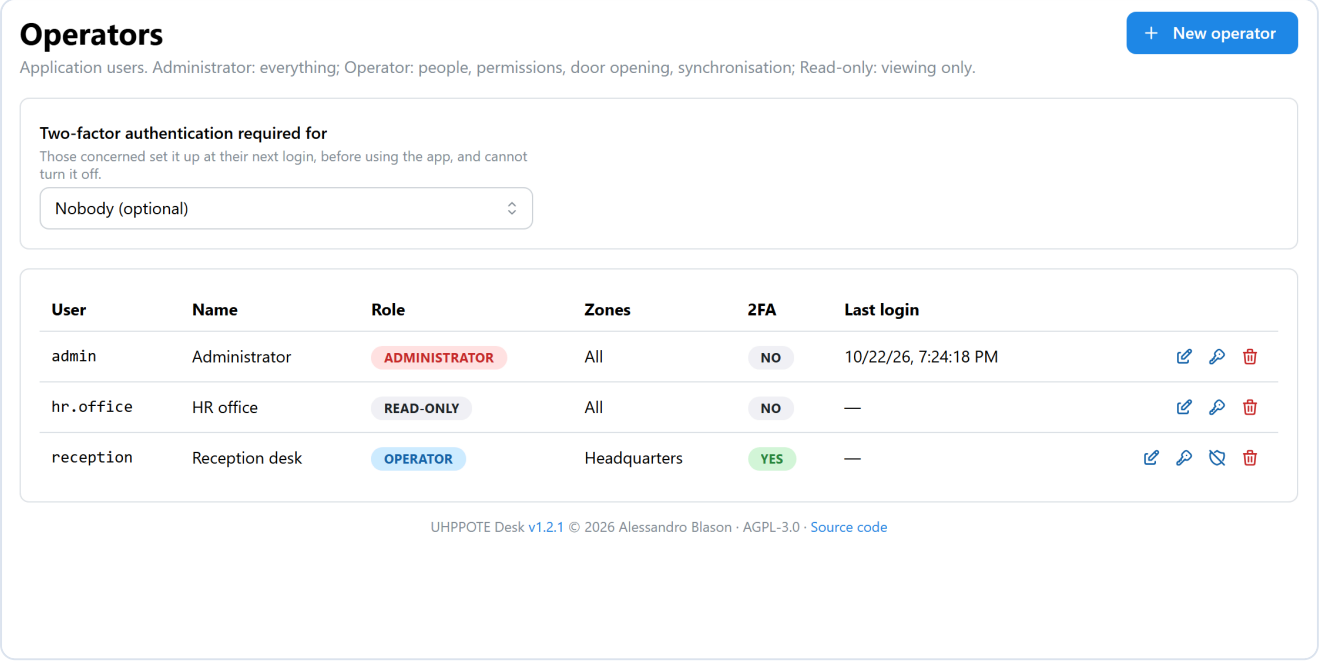



**Figure 35** — Operators of the application.

The accounts that can log in to the application. For each operator: username, display name, role (see chapter 4), zones and last login.

- New operator — choose username, name, role and optionally zones. If you leave the password empty, a temporary password is generated and shown once: give it to the person, who must change it at first login.
- **Reset password** — generates a new temporary password.
- **Active** — a disabled operator cannot log in; their operations remain in the audit log, as they do when an operator is deleted.

You cannot lower your own role, disable or delete yourself, so there is always at least one administrator.

- **Two-factor authentication required for** — nobody (optional), administrators, operators and administrators, or everybody. Those concerned must set it up at their next login, before using the application, and cannot turn it off. The change is recorded in the audit log.
- **2FA** column — who has two-factor authentication active. The **crossed shield** icon resets it for another operator who lost the phone: they log in with the password only and can set it up again; their sessions and remembered devices are closed.

ADMINISTRATION

# 23 · Audit log

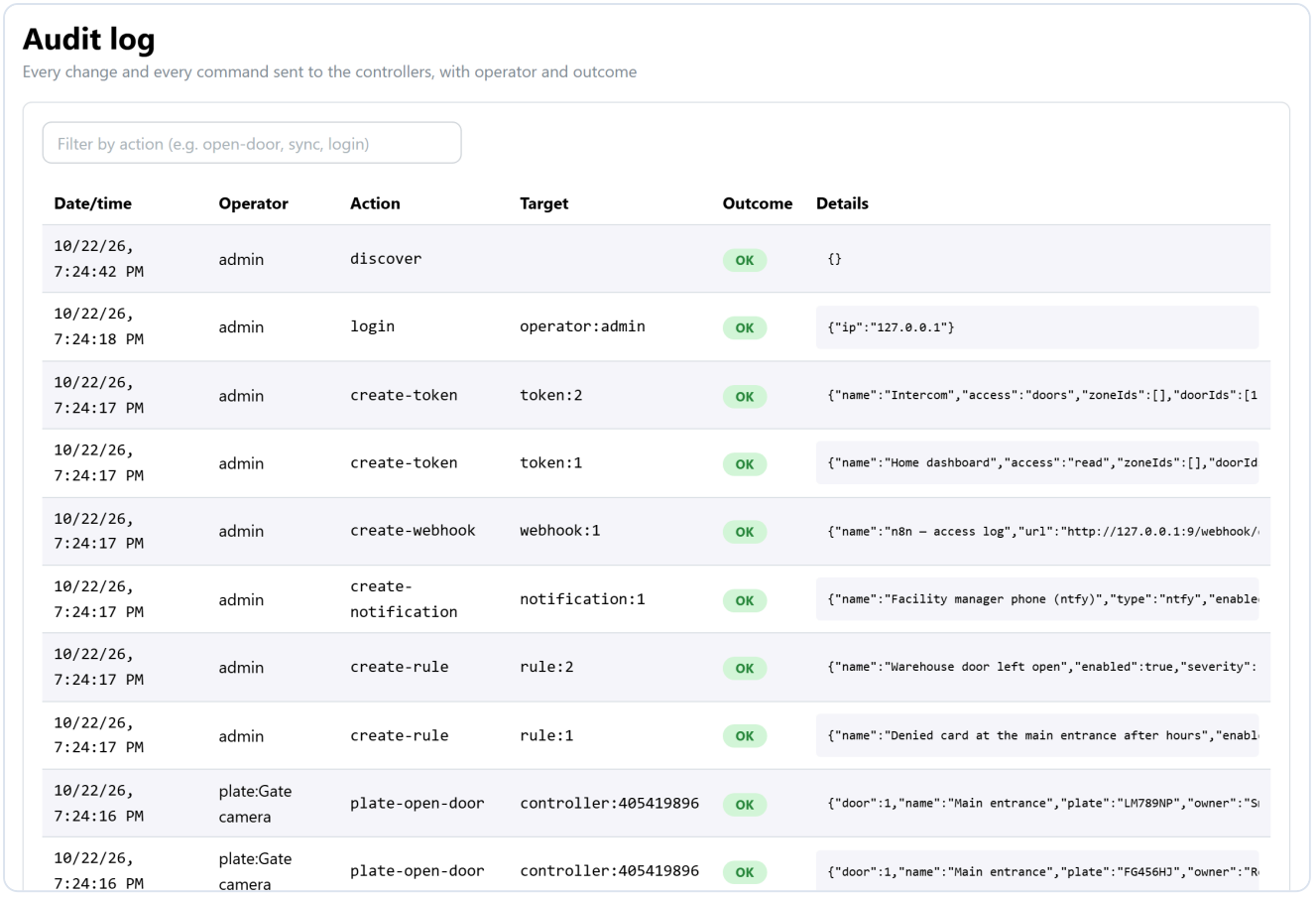



**Figure 36** — Every change and every command, with operator and outcome.

The log of every change made in the application and every command sent to the controllers: date and time, operator (or *system* for automatic operations), action (e.g. `open-door`, `put-card`, `sync-cards`, `login`), target, outcome and details. Filter by action to answer questions like "who opened the back office door yesterday?".

ADMINISTRATION

# 24 · System

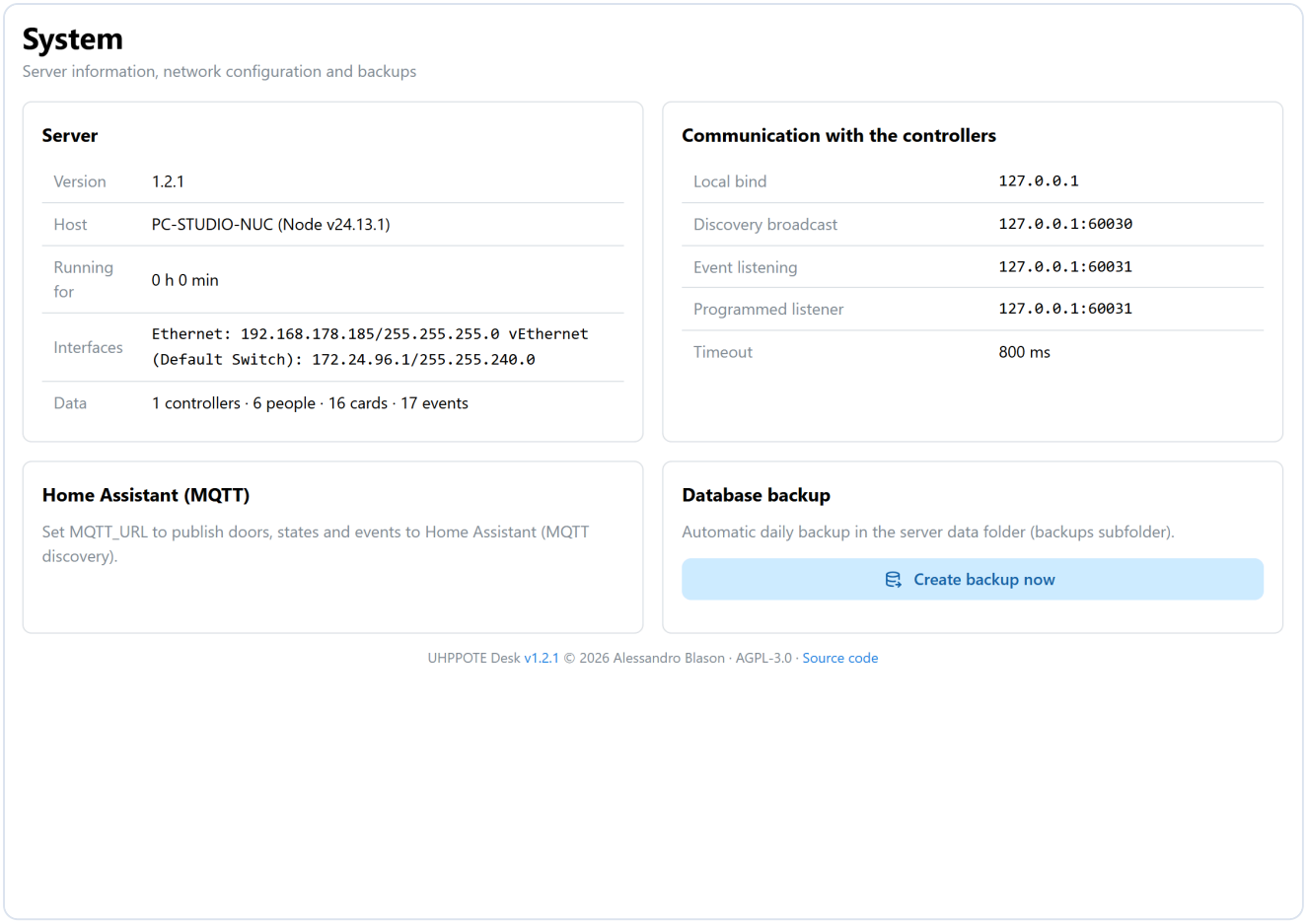



**Figure 37** — Server information.

- **Server** — version, host name, uptime, network interfaces, and the amount of data (controllers, people, cards, events).
- **Communication with the controllers** — the network settings of the server: local address, discovery broadcast, event listening port and the **programmed listener**, i.e. the address the controllers send their events to. A warning appears if it is not set: events then arrive only with the periodic download, not in real time.
- **Database backup** — a copy of the database is made every night (at 02:00 by default, the last 14 are kept) in the `backups` folder of the server data. **Create backup now** makes one immediately, e.g. before a large import. Administrators only.

## 25 · Change password

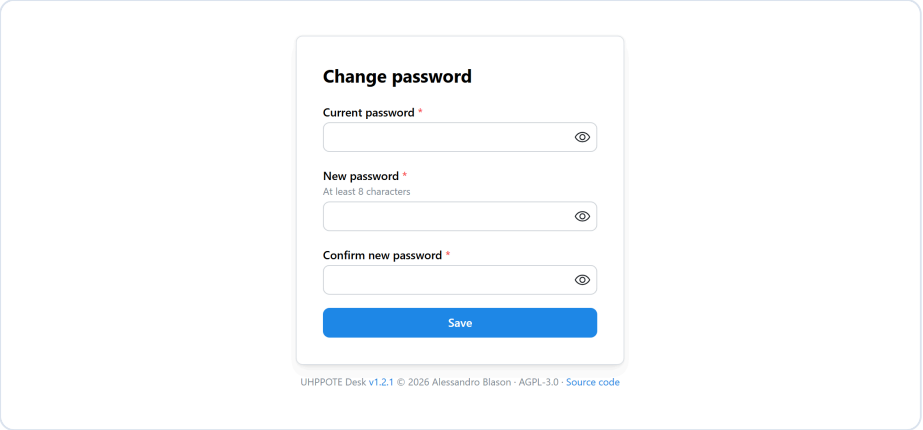



**Figure 38** — Changing your password.

From the key icon in the header. Enter the current password and the new one twice (at least 8 characters). Your other open sessions are closed. If the `admin` password is set by the server's `ADMIN_PASSWORD` variable, the page explains that it can be changed only there.

YOUR ACCOUNT

## 26 · Two-factor authentication

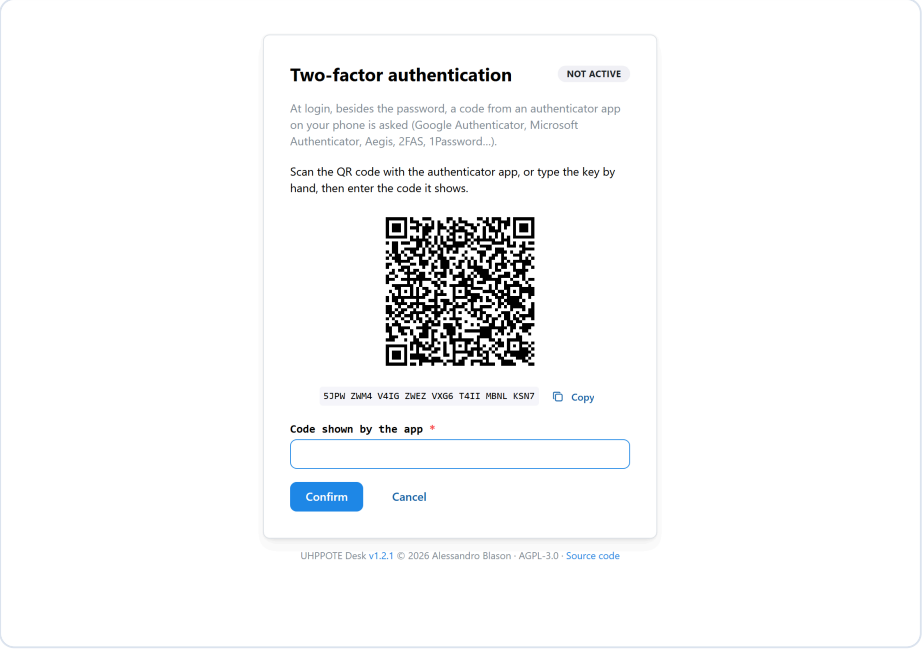



**Figure 39** — Activating two-factor authentication: scan the QR code with an authenticator app.

Besides the password, the login asks for a 6-digit code from an authenticator app on your phone (Google Authenticator, Microsoft Authenticator, Aegis, 2FAS, 1Password...). Every operator activates it from the **shield icon** in the header (green when active):

1. Activate and confirm your password.
2. Scan the QR code with the app, or type the key shown under it, then enter the code the app shows and Confirm.
3. Save the **10 recovery codes** (copy or download them) and keep them away from the phone: each one replaces the app code once, and they are not shown again.

From then on the login asks for the code after the password; the other open sessions and remembered devices (mobile app) are logged out and log in again with the code. A code is accepted once, and wrong codes count as failed login attempts. On the same page you can generate **new recovery codes** or **turn it off** (password and code).

**Lost phone.** Log in with a recovery code; or another administrator resets it from the Operators page.

> **Emergency access for administrators**
> If the only administrator has lost both the phone and the recovery codes, set `ADMIN_2FA_BYPASS=true` on the server (Home Assistant add-on: option *Emergency access without two-factor authentication*) and restart. Administrators then log in with the password only and can turn their two-factor authentication off without a code, also when it is mandatory (it is then set up again with the new phone at the next login). Operators and read-only users are not concerned: they still need their code. While it is set, a red banner is shown on every page and every such login is marked in the audit log. Remove the variable and restart as soon as access is restored.

PART 6

# Automation and integrations

Rules, notifications, webhooks, API tokens, the mobile app and Home Assistant. Administrators only.

ADMINISTRATION

## 27 · Rules

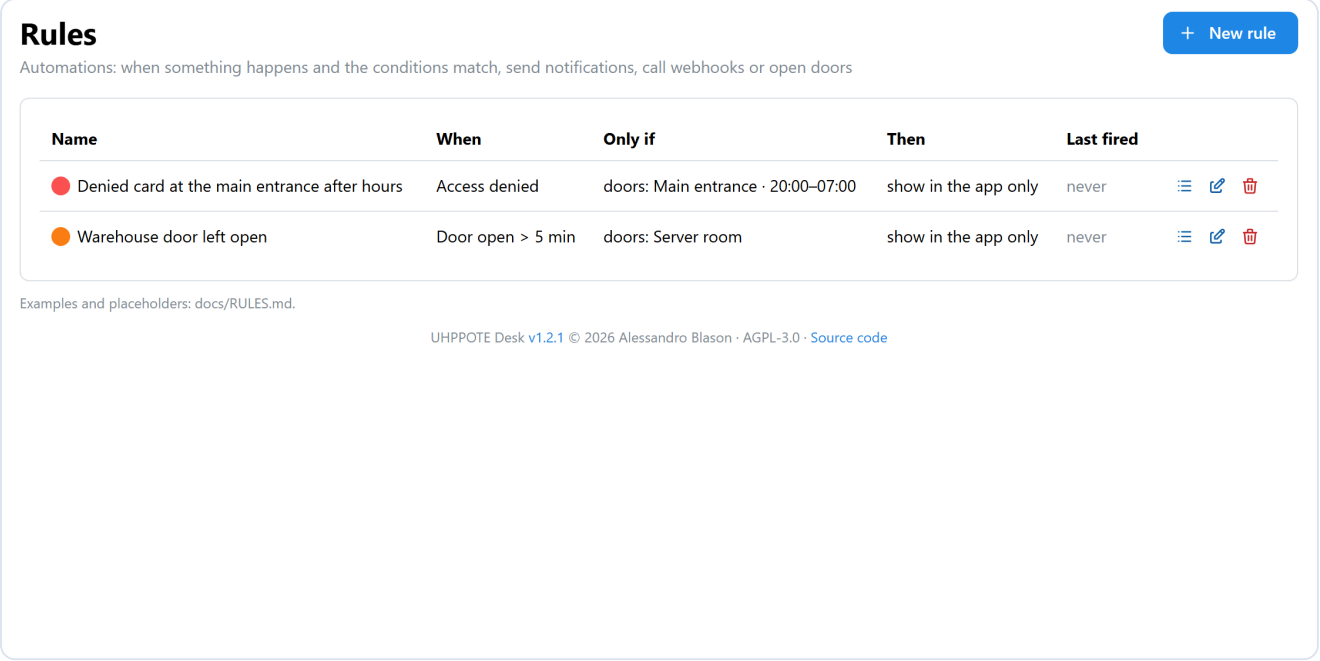



**Figure 40** — Automation rules.

A rule says **when** something happens, **only if** some conditions match, **then** do something. For example: a denied card at the main entrance after 20:00 → alarm notification; the warehouse door open for more than 5 minutes → warning.

- **When** — any event (access granted or denied, door event, alarm, door opened or closed, controller offline or back online, unusual time, repeated denied attempts, guest link, licence plate) or **door open longer than** a number of seconds (needs the door sensor).
- **Only if** — doors, people or cards, a time window (it may cross midnight) and days of the week.
- **Then** — send a notification to some channels, call some webhooks, open some doors. Every firing is also shown in the web app: alarms stay on screen until closed.
- **Severity** (information, warning, alarm) sets emoji and priority of the notifications; the **message** may use `{person}`, `{card}`, `{door}`, `{time}`…; the **pause** avoids repeated firings.

ADMINISTRATION

# 28 · Notifications

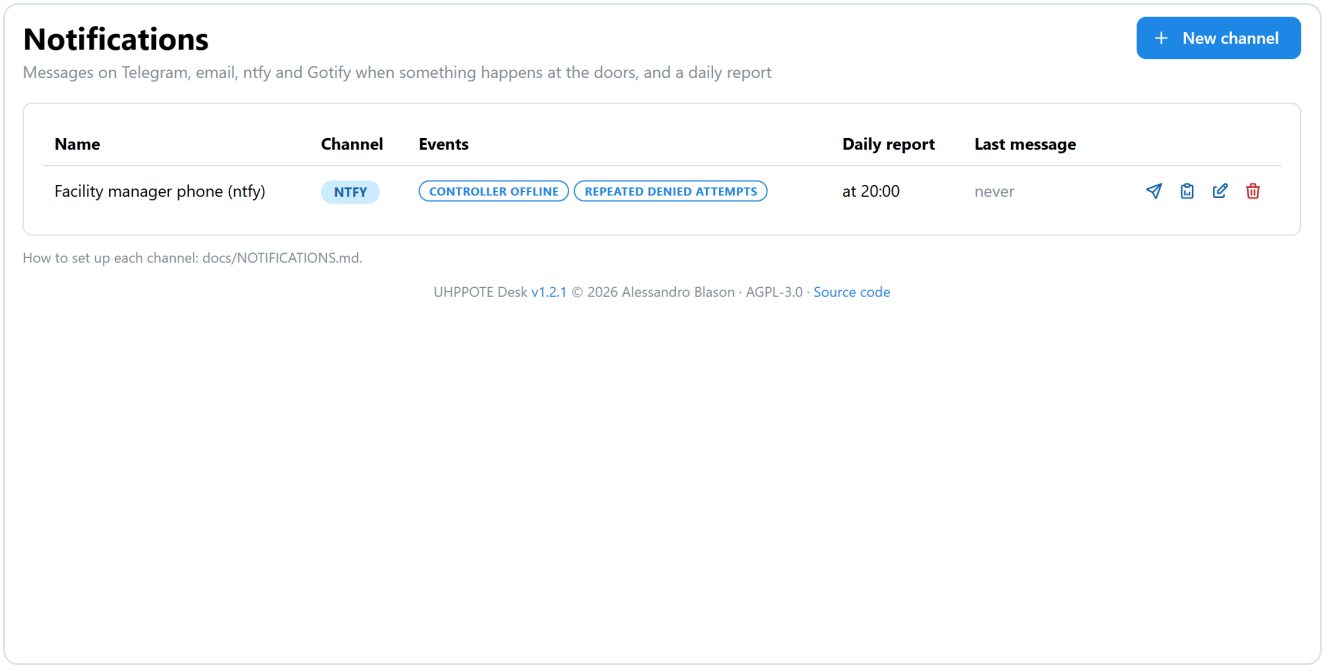



**Figure 41** — Notification channels.

Messages on **Telegram**, by **email**, or to **ntfy** and **Gotify** (push servers that can run on your own NAS). Each channel has the events that send a message, an optional door filter, the language and the optional **daily report** (who came in, denied accesses, alarms and anomalies of the last 24 hours). On Telegram the messages about a door can carry an **Open door** button, usable only by the Telegram users listed in the channel and for 15 minutes. Send test checks the channel at once. Setup of each type: `docs/NOTIFICATIONS.md`.

ADMINISTRATION

# 29 · Webhooks

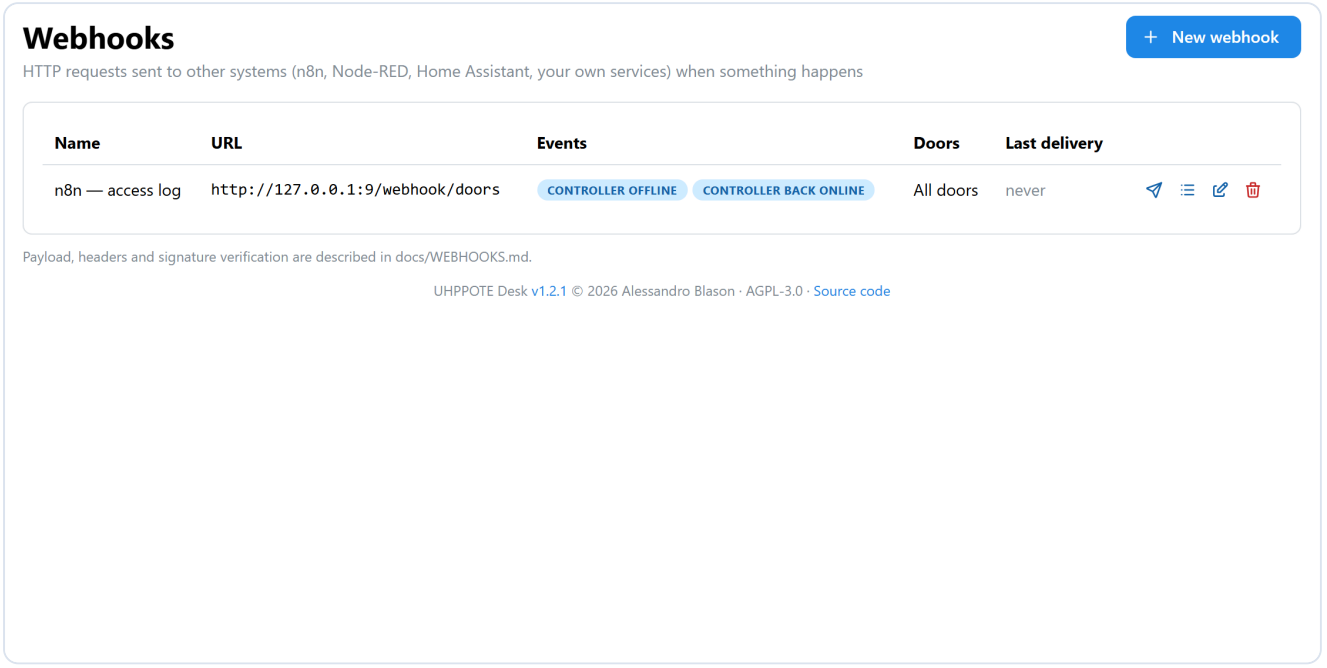



**Figure 42** — Outgoing webhooks.

An HTTP `POST` request to other systems (n8n, Node-RED, Make, a Home Assistant webhook trigger, your own service) when something happens: the URL, the events, an optional door filter and extra headers. Every request is signed with the webhook's secret ( `X-UHPPOTE-Signature` ). **Send test** sends a test request; **Deliveries** lists the last 100 requests with status code and payload. Payloads and signature: `docs/WEBHOOKS.md` .

ADMINISTRATION

## 30 · API tokens

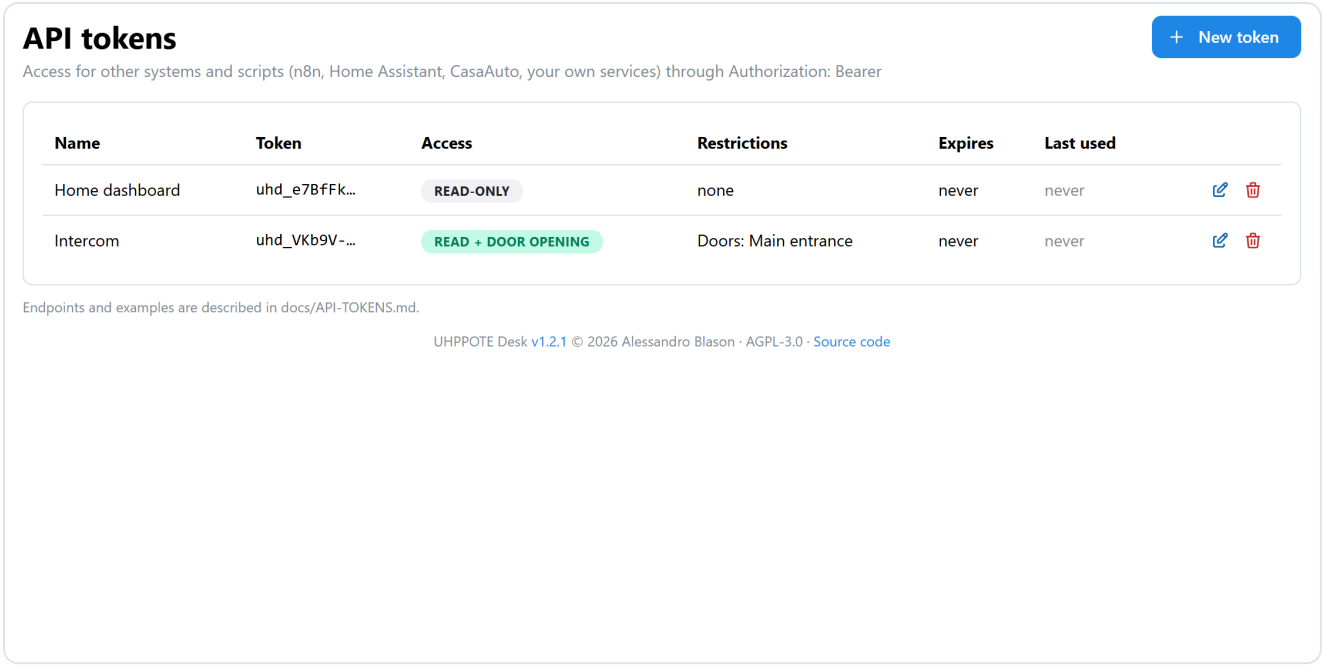



**Figure 43** — API tokens for other systems and scripts.

A token lets another system use the UHPPOTE Desk API without an operator login: a home automation opening the gate, a script reading events, an HR system creating people. Access levels: **read-only**, **read + door opening**, **operator**, **administrator**; optionally restricted to some zones or to the doors it may open, and with an expiry date. The token ( `uhd_…` ) is shown only once; disable or delete it to revoke it. Everything done with a token is in the audit log as `token:<name>` . Endpoints: `docs/API-TOKENS.md` .

ADMINISTRATION

# 31 · Mobile app

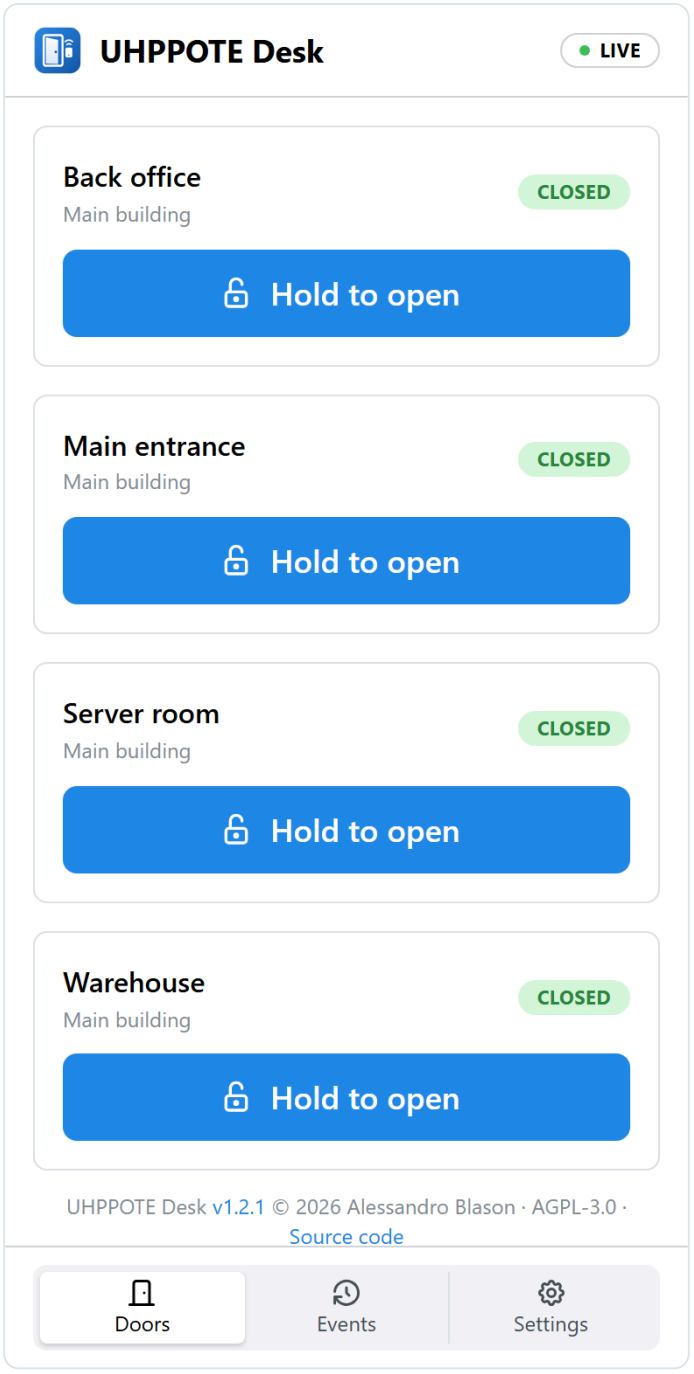



**Figure 44** — The mobile app on a phone.

The page `/m` is made for phones and can be installed on the home screen like an app. It shows the doors with their live state, opens them with a confirmation, lists the latest events and receives **push notifications**. The phone icon in the header opens it.

- **Remember this device** at login gives the phone its own long login: no expiry with fingerprint/Face ID registered, otherwise 30–180 days.
- Door opening is confirmed by **holding the button** or, if the administrator chooses it, with the phone's **fingerprint/Face ID**.
- Installation, fingerprint/Face ID and push notifications need HTTPS ( `PUBLIC_URL` and a reverse proxy).

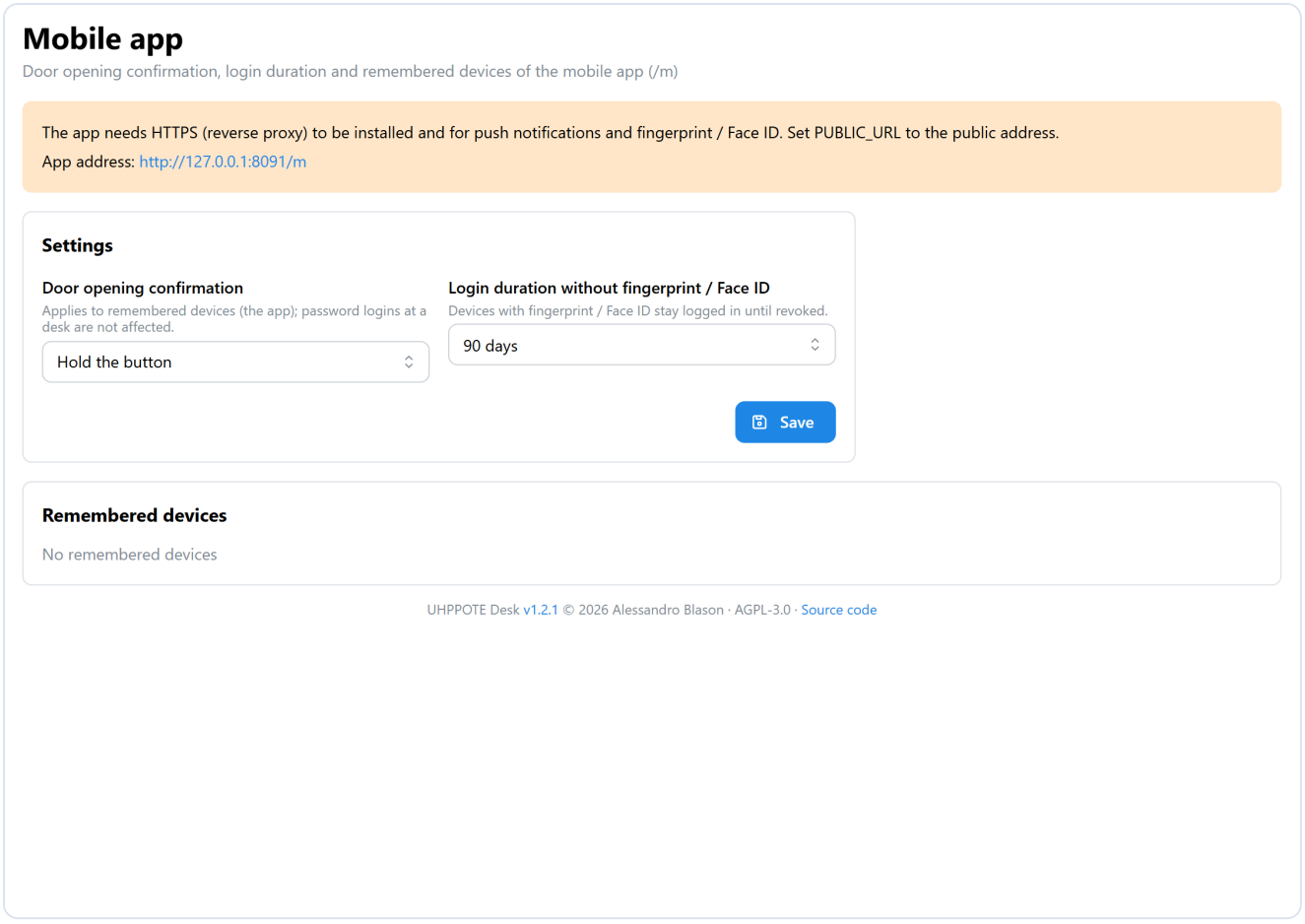



**Figure 45** — Mobile app settings and remembered devices.

In Administration → Mobile app administrators choose the opening confirmation and the login duration, and see every remembered device (operator, last use, fingerprint/Face ID, push) to revoke it. Details: `docs/MOBILE.md`.

## 32 · Home Assistant

UHPPOTE Desk publishes its controllers and doors to an MQTT broker with **Home Assistant MQTT discovery**: the devices appear in Home Assistant by themselves. On Home Assistant OS the simplest way is the **UHPPOTE Desk add-on** (repository `https://github.com/alexbn71/UHPPOTE-Desk`): with the Mosquitto broker add-on the connection is configured automatically. With Docker, set `MQTT_URL`, `MQTT_USERNAME` and `MQTT_PASSWORD`; the status is on the System page.

| Entity | Meaning |
|---|---|
| **Controller** | |
| Connectivity, Problem | controller reachable; system error reported |
| Fire alarm | fire alarm active: on at once at the fire event, off a few seconds after the input is released |
| Forced lock | forced lock active |
| Expansion output 1–4 | outputs of the UT0311-01A board (only with *Expansion board connected*) |
| **Door** (one per active door) | |

| Entity | Meaning |
|---|---|
| Door, Lock | door sensor (on = open); relay (on = unlocked) |
| Open | button that opens the door (can be disabled with `MQTT_DOOR_CONTROL=false`) |
| Access | event entity: granted, denied, door, alarm — with person, card, direction and reason |
| Last access | name of the last person who swiped |

Entity names follow `MQTT_LANGUAGE` (add-on option *MQTT language*). Example automations: `docs/HOME-ASSISTANT.md`.

PART 7

# Reference

Event reasons, door modes and rules, troubleshooting and glossary.

## 33 · Event reasons

Every event carries a reason code, shown as text in the monitor and the history.

| Code | Reason | Meaning |
|---|---|---|
| 1 | Card swipe | access granted with a card |
| 6 | No access rights | card unknown to the controller, or no permission on that door |
| 7 | Incorrect password | wrong PIN on the keypad |
| 8 | Anti-passback | the card must first pass through a reader of the other group |
| 9 | Multi-card | more cards are required to open |
| 10 | First card | door waiting for the first authorised card |
| 11 | Door normally closed | the door is set to normally closed |
| 12 | Interlock | another door of the interlock group is open |
| 13 | Outside time profile | the permission is not valid at this day/time |
| 15 | Invalid time zone | the time profile is not defined on the controller |
| 18 | Access denied | access refused by the controller for a reason not covered by the other codes |
| 20 | Exit button | door opened with the exit button |
| 23 / 24 | Door opened / Door closed | door sensor (with "Record door and button events") |
| 25 | Door opened (supervisor password) | opened with a door passcode |
| 28 / 29 | Controller power on / reset | the controller was switched on or reset |
| 31–34 | Button invalid | exit button refused (door locked, offline, interlock, threat) |
| 37 | Door open too long | the door was left open |
| 38 | Forced open | door opened without a valid access |
| 39 | Fire | fire alarm input active |
| 40 | Forced closed | forced lock input active |

| Code | Reason | Meaning |
|---|---|---|
| 41–43 | Theft prevention · 24×7 zone · Emergency | alarm inputs |
| 44 / 45 | Remote opening | opened from the application (or a USB reader) |

Rows of denied accesses and alarms are highlighted in red in the dashboard, monitor and history.

## 34 · Door modes and access rules

### Door modes

| Mode | Behaviour |
|---|---|
| Controlled | normal operation: the lock opens for an authorised card, the exit button or a remote opening |
| Normally open | the lock is always released (e.g. during opening hours, with a scheduled task) |
| Normally closed | the lock is always locked; cards are refused (reason 11) |

### Interlock

| Option | Doors that cannot be open at the same time |
|---|---|
| Doors 1 and 2 | 1 with 2 |
| Doors 3 and 4 | 3 with 4 |
| Doors 1-2 and 3-4 | 1 with 2, and 3 with 4 (two independent airlocks) |
| Doors 1, 2 and 3 | any two of 1, 2 and 3 |
| Doors 1, 2, 3 and 4 | any two of the four doors |

### Anti-passback

| Option | Reader groups (entry : exit) |
|---|---|
| (1:2);(3:4) | reader 1 alternates with reader 2, and reader 3 with reader 4 |
| (1,3):(2,4) | readers 1 and 3 alternate with readers 2 and 4 |
| 1:(2,3) | reader 1 alternates with readers 2 and 3 |
| 1:(2,3,4) | reader 1 alternates with readers 2, 3 and 4 |

On a 4-door controller with one reader per door, "reader N" is the reader of door N. On 1- and 2-door controllers with in and out readers, check the controller documentation for the numbering.

## 35 · Troubleshooting

| Problem | What to check |
|---|---|
| The network search finds nothing | Server and controllers on the same network segment; cable and power; firewall allowing UDP port 60000; with Docker, "host" networking. Try adding the controller manually with its IP address. |
| Controller shown Offline | IP address and port in *Network and configuration*; ping the controller; another program using the controller at the same time. |
| Swipes do not appear in the Monitor, but appear later in the history | Real-time events are not reaching the server: set `UHPPOTE_LISTENER_PUBLIC` on the server (see the warning on the System page), allow inbound UDP port 60001, then *Configure automatically* the listener on the controller page. |
| A card is refused although it has permissions | Look at the reason in the history: 13 means outside the time profile; 6 means the card is not on the controller or has no permission on that door — run *Compare* on the Synchronisation page. Check that the person and the card are active and that today is within the person's validity period. |
| "Saved, but not sent to every controller" | Some controller was offline. When it is back, use *Align cards* or *Full synchronisation*. |
| "Card number not valid for Wiegand-26" | The number must have facility ≤ 255 and code ≤ 65535. Swipe the card and use the number shown among the unknown cards. |
| Wrong time in events or reports | Server time zone ( `TZ` variable); the controller clock is corrected automatically, or use *Synchronise time*. |
| An expansion board output does not switch | Use *Read from the controller* to see what the controller stores; check that *Expansion board connected* is on and save again; that the output is active, with the right door and events; that the alarm really happens (the event appears in the Monitor: 37 open too long, 38 forced open, 39 fire). Forced open needs a door sensor; duress needs a PIN keypad. |
| Forgotten administrator password | Set `ADMIN_PASSWORD` on the server and restart it; then log in as `admin`. |
| Lost the phone with the authenticator app | Log in with a recovery code; or another administrator resets it (*Operators*, crossed shield). The only administrator: `ADMIN_2FA_BYPASS=true` , restart, turn it off on the *Two-factor authentication* page, remove the variable and restart (see Two-factor authentication). |
| "Verification code not valid" with the right app | The clocks of the server and of the phone must be correct (NTP): codes change every 30 seconds. |
| The expansion outputs do not appear in Home Assistant | *Expansion board connected* must be on in the *Expansion board* tab; save, and the entities *Expansion output 1–4* are created. |

## 36 · Glossary

| Term | Meaning |
|---|---|
| Broadcast | a network message sent to all the devices of the local network; used to find controllers |
| Controller | the access control board (UHPPOTE UT0311-L01/L02/L04) |
| Event index | progressive number of an event in the controller memory |

| Term | Meaning |
|---|---|
| Expansion board | the UT0311-01A add-on board with four alarm relays and a fire input |
| Guest link | a link that opens some doors for a while, without card or app |
| Two-factor authentication (2FA) | a code from an authenticator app asked at login besides the password |
| Recovery code | a single-use code that replaces the app code when the phone is lost |
| Facility code | the first part (0–255) of a Wiegand-26 card number |
| First card | function that keeps a door locked until an authorised person arrives |
| Interlock | doors that cannot be open at the same time (airlock) |
| Anti-passback | prevents a card from entering twice without exiting |
| Listener | the server address to which a controller sends its events |
| Punch | a swipe used for attendance (clock in / clock out) |
| Relay | the output of the controller that releases the electric lock |
| Synchronisation | aligning a controller with the application database |
| Time profile | days and hours in which a permission is valid (ID 2–254) |
| Wiegand-26 | the card format accepted by the controllers: facility × 100000 + code |
| Zone | group of controllers, used to restrict operators |